



**Quantum-Enhanced CNN for Highly
Accurate Breast Cancer Classification:
A Proof-of-Concept Study**

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Abstract

Breast cancer is one of the most common cancers in women worldwide, and while deep learning models like CNNs have shown strong results in automating tissue image analysis, they still struggle with real-world data complexity and are hitting performance ceilings. This thesis tests whether adding a quantum layer on top of a classical CNN can produce real, measurable improvements in breast cancer classification, using the BreakHis dataset of 7,909 high-resolution microscopy images across 8 tumor subtypes. Through a four-stage experiment covering classical model selection, quantum configuration search, statistical analysis, and direct comparison using McNemar's test, the best classical model (regnetx_002) reached 90.80% accuracy, while the best quantum hyper-parameter combination, tested on the model mobilenetv3_small_100, including SimplifiedTwoDesign quantum circuit with 8 qubits and 3 layers, reached 92.72%, a 6.42 percentage point gain over its classical counterpart. The quantum layer improved accuracy significantly across all seven tested backbones ($p < 0.001$) and boosted cancer detection recall across all seven models, with three of those cases achieving statistical significance. The maximum overall gains reached 18.35 percentage points for accuracy and 2.72 percentage points for recall. Interestingly, choosing the right circuit design mattered much more than using more qubits, with 58.3% versus 16.7% of comparisons showing statistically significant differences, respectively. Overall, these results show that quantum-enhanced CNNs can outperform classical models on realistic medical imaging data, though future work is still needed to test these findings on real quantum hardware and fine-tune quantum settings for each backbone individually.

Keywords

Breast Cancer Classification, Histopathology, Quantum Machine Learning, Convolutional Neural Networks, Hybrid Quantum-Classical Models

Supervisor

Asso Prof. Arif Ali Khan

Foreword

This thesis has been a very special journey for me. I would like to begin by expressing my gratitude to my thesis supervisor, Professor Arif Ali Khan, for all the support he gave me throughout this process. His thoughts and advice were always helpful when I confronted an obstacle. I could learn from him not only about the thesis topic, but also how to work professionally and communicate in a research environment, which was a treasured opportunity for me.

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With heartfelt gratitude,

Sang Vo

Oulu, June 2026

Declaration of Generative AI in the Thesis

During this work, I used Grammarly and ChatGPT only to check and correct any grammatical error in my writings, thereby improving the quality and clarity of this thesis. I confirm that I did not utilize Generative AI to directly write and perform any analysis on my behalf. All study design, analysis, and literature review conducted during the process were implemented and completed by myself, without any usage of AI.

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Abbreviations

A	Adenosis
AI	Artificial Intelligence
BreakHis	Breast Cancer Histopathological Image Classification
CAD	Computer-Aided Diagnosis
CNN	Convolutional Neural Network
CNOT	Controlled NOT
DC	Ductal Carcinoma
DP	Digital Pathology
F	Fibroadenoma
GPU	Graphics Processing Unit
H&E	Haematoxylin and Eosin
HOG	Histogram of Oriented Gradients
LC	Lobular Carcinoma
MC	Mucinous Carcinoma
NAS	Neural Architecture Search
NISQ	Noisy Intermediate-Scale Quantum
PC	Papillary Carcinoma
PNG	Portable Network Graphics
PT	Phyllodes Tumor
QCNN	Quantum Convolutional Neural Network
QML	Quantum Machine Learning
QNN	Quantum Neural Network
RAM	Random Access Memory
ReLU	Rectified Linear Unit
RGB	Red, Green, Blue
RQ	Research Question
RX	Rotation-X
SIFT	Scale-Invariant Feature Transform
SVM	Support Vector Machine
TA	Tubular Adenoma
VQC	Variational Quantum Circuit

1 Introduction

This chapter establishes the context for the research presented in this thesis. We begin by looking at why accurate breast cancer diagnosis matters and where current deep learning approaches still fall short when dealing with real-world medical images. From there, we introduce quantum computing as a potential way forward and explain why hybrid quantum-classical models have been attracting growing research interest. We then narrow the focus to the specific gap this thesis addresses, before closing with a clear statement of the objectives and research questions that guide the rest of the work.

1.1 Current Situation in Breast Cancer Diagnosis

The process of diagnosing breast cancer is currently facing many challenges, from high patient numbers to the limits of current technology. This section explains the main problems in the healthcare system and looks at why traditional AI models are no longer enough to solve them.

1.1.1 Clinical Urgency and the Diagnostic Bottleneck

Breast cancer remains the most frequently diagnosed malignancy and the leading cause of cancer-related mortality among women globally. In 2020 alone, an estimated 2.3 million new cases were identified, and projections indicate a 40% increase in the global burden by 2040 (Sung et al., 2021). Despite advancements in screening, the gold standard for definitive diagnosis remains the histopathological examination of biopsy tissue.

However, this critical diagnostic step is currently facing two compounding challenges. First, there is a significant global shortage of trained pathologists, leading to increased workloads and burnout (Metter et al., 2019). Second, manual tissue analysis is inherently subjective. Even among experts, there is documented inter-observer variability, where the interpretation of complex morphological features can lead to different diagnostic conclusions for the same patient (Elmore et al., 2015). These human bottlenecks create an urgent need for automated, highly reliable assistance to prevent false negatives - which delay life-saving treatment - and false positives, which trigger unnecessary patient anxiety and invasive follow-ups.

1.1.2 The Role and Limitations of Classical AI

To mitigate these challenges, Digital Pathology (DP) has increasingly turned to Computer-Aided Diagnosis (CAD) systems. Convolutional Neural Networks (CNNs) have become the standard in this space, demonstrating a remarkable ability to automate feature extraction and identify patterns in high-resolution tissue slides (Huang et al., 2017; Simonyan & Zisserman, 2015; Tan & Le, 2021). By serving as a decision-support tool, these systems help pathologists handle higher volumes with greater consistency.

Despite their success in controlled benchmarks, classical CNNs face significant hurdles in real-world clinical deployment:

- **Data Variability:** Real histopathology is complicated. Variations in tissue preparation, staining protocols, and artifacts often lead to "domain shift," where a model trained on one dataset fails to generalize to the subtle variability of another hospital's slides.
- **The Computational Wall:** Achieving incremental gains in diagnostic accuracy has traditionally required scaling up model depth and parameter counts. This resource-intensive scaling approach is reaching a point of diminishing returns. The energy consumption and specialized hardware required to train and run these massive models are becoming unsustainable, creating a practical limit on how much "smarter" classical hardware can get (Dhar et al., 2023; Thompson et al., 2020).

As classical architectures begin to plateau, it becomes clear that simply adding more layers is no longer a sufficient solution. This situation necessitates the exploration of new computational paradigms that can process high-dimensional medical data more efficiently. Quantum computing, with its ability to operate in exponentially large Hilbert spaces, offers a theoretically grounded path to capturing the complex morphological patterns in breast tumor tissue that classical feature spaces struggle to represent efficiently.

1.2 The Emergence of Quantum Computing

To overcome the limits of today's technology, researchers are looking at quantum computing as a new way to process information. This section explains the basic rules of quantum systems and looks at how they can be used to improve artificial intelligence and medical research.

1.2.1 A New Computational Paradigm

To address the looming computational wall of classical hardware, researchers have increasingly turned toward quantum computing as a fundamental shift in information processing. While classical computing is constrained by the binary nature of bits - which exist strictly as 0 or 1 - quantum computing leverages the principles of quantum mechanics. The primary unit of information, the qubit, utilizes **superposition** to exist in multiple states simultaneously and **entanglement** to create complex correlations between qubits that have no classical analogue.

These properties allow quantum computers to perform certain types of calculations at scales that would be intractable for even the most powerful classical supercomputers. Specifically, in the context of machine learning, quantum systems can process information in a high-dimensional **Hilbert space**. This provides a mathematical framework to represent and analyze data in ways that classical feature spaces cannot easily replicate (Preskill, 2018).

1.2.2 Quantum Machine Learning and Feature Mapping

The intersection of these quantum principles with Artificial Intelligence has birthed the field of Quantum Machine Learning (QML). One of the most compelling arguments for QML is the concept of **quantum-enhanced feature spaces**. By mapping classical data

into the vast state space of a quantum system, it is possible to identify non-linear patterns and "hidden" structures that are computationally expensive for classical kernels to detect (Havlíček et al., 2019; Schuld & Killoran, 2019).

In the current "Noisy Intermediate-Scale Quantum" (NISQ) era, where fully error-corrected quantum computers are not yet available, **Variational Quantum Circuits (VQCs)** have emerged as the leading tool for practical application. VQCs are flexible, trainable circuits that can be optimized using classical gradient-based methods, making them ideal candidates for integration into existing deep learning pipelines (Cerezo, Arrasmith, et al., 2021; Mitarai et al., 2018).

1.2.3 The Hybrid Path for Medical Imaging

Given the current limitations of quantum hardware, the most viable path forward is the development of **hybrid quantum-classical architectures**. In this setup, a classical model (like a CNN) handles the task of processing high-resolution raw pixels, while a quantum layer acts as a specialised "feature processor" or a "classifier". Early proof-of-concept studies demonstrate that quantum layers can capture complex, high-dimensional patterns, often achieving competitive or superior performance in binary classification scenarios where training data is limited (Matondo-Mvula & Elleithy, 2024; Rifat et al., 2024; Tasnim et al., 2024). Whether these results hold on larger, more realistic datasets remains an open question that this thesis directly addresses.

1.3 Research Gaps and Motivation

Given the current limitations of NISQ-era hardware, the hybrid quantum-classical architecture has emerged as the most viable path for practical application. In this setup, a classical model such as a CNN handles the processing of high-resolution raw pixels, while a quantum layer acts as a specialised feature processor or classifier. Early proof-of-concept studies have shown that quantum layers can capture complex, high-dimensional patterns, often achieving competitive or superior performance in binary classification scenarios where training data is limited (Matondo-Mvula & Elleithy, 2024; Rifat et al., 2024; Tasnim et al., 2024). These results suggest a genuine potential for quantum-enhanced feature spaces to complement classical deep learning in medical imaging tasks.

However, the current body of research leaves several critical questions unanswered. Much of the existing work on hybrid models for breast cancer detection relies on simplified experimental setups that do not reflect clinical reality. Prior studies utilise datasets such as BreastMNIST, which consists of low-resolution, simplified versions of medical images, or work with small subsets of larger datasets. These idealised environments discard high-frequency details, staining variations, and morphological complexities inherent in real-world histopathology. Without testing on high-resolution and sufficiently large data, it is impossible to determine whether reported performance gains are genuine or merely an artefact of working with reduced data dimensions.

Furthermore, many hybrid studies leave significant structural questions unanswered. Researchers often fail to provide a clear technical justification for their design choices, such as why a specific classical backbone was selected or how key quantum parameters like qubit count, circuit depth, and encoding method were determined. This creates a lack

of interpretability, leaving a gap in our understanding of how these components truly work together and whether the quantum layer provides a unique computational benefit or simply acts as a redundant noise filter. The absence of rigorous statistical validation compounds this problem further, as most studies do not apply formal hypothesis testing to determine whether observed performance differences are statistically significant or within the margin of chance.

Taken together, these gaps point to a clear motivation for this research. Classical CNNs are approaching a computational ceiling in histopathological classification. Hybrid quantum-classical models have demonstrated early promise but have not been stress-tested on realistic, high-resolution data under controlled and reproducible conditions. The field is at a point where a systematic, statistically rigorous study is needed to determine whether the theoretical advantages of quantum feature spaces translate into measurable clinical value.

As a result, several questions remain open. When moving beyond simplified datasets to full, complex histopathological image collections, the following challenges demand systematic investigation:

- What classical CNN architecture performs best for breast cancer classification under a predefined constraint of computational resources?
- How do key quantum circuit parameters, such as qubit count, circuit depth, and circuit template, influence classification performance in a hybrid model?
- Does adding a quantum component actually lead to an improvement over purely classical models?

1.4 Thesis Objectives and Contributions

This thesis aims to answer all the open questions stated in Section 1.3 by conducting a systematic comparison between hybrid quantum-classical neural networks and purely classical baselines. The study is carried out using the Breast Cancer Histopathological Image Classification (BreakHis) dataset, which provides a more challenging and realistic set of tumour tissue images than many simplified alternatives (Spanhol et al., 2015). To address each question in turn, the research is structured around four central research questions:

RQ₁

What is the best classical CNN backbone for the task of breast cancer image classification?

Rationale: To evaluate a hybrid model fairly, it must be compared against a high-performing classical baseline rather than a weak or sub-optimal one. Answering this question establishes a rigorous benchmark that ensures any observed quantum advantage is measured against a state-of-the-art classical foundation.

RQ₂

What is the optimal hyperparameter configuration for the quantum layer?

Rationale: Variational quantum circuits are highly sensitive to their specific architectural and hyperparameter settings. Answering this question ensures the hybrid model is evaluated using the best possible quantum configuration to accurately reflect its true performance potential.

RQ₃

Which specific hyperparameters significantly influence the classification accuracy of the hybrid model?

Rationale: Beyond simply finding the best settings, it is important to identify which specific design choices drive success and which do not. Answering this question helps us understand how these quantum components interact with each other and how they directly impact the classification performance.

RQ₄

Does the inclusion of a quantum layer improve the classification performance compared to classical approaches?

Rationale: The central hypothesis of this thesis is that quantum-enhanced feature mapping can outperform purely classical methods when applied to breast cancer histopathology images. Answering this question determines whether the integration of a quantum layer provides a statistically significant improvement in diagnostic accuracy for this specific medical task.

The thesis is expected to contribute (1) a reproducible classical baseline benchmark on BreakHis under realistic resource constraints, (2) an empirical analysis of quantum circuit hyperparameter sensitivity, and (3) a statistically grounded evaluation of whether hybrid quantum-classical models offer a genuine advantage over their classical counterparts in real-world histopathological classification.

2 Background

Building a robust hybrid quantum-classical model for medical image classification requires knowledge from two distinct but complementary bodies of knowledge. This chapter lays out the theoretical background that supports the experimental work presented later in this thesis. We begin with classical Convolutional Neural Networks, which form the backbone of our image processing pipeline, examining how they extract hierarchical features from raw pixels. We then shift to the fundamentals of quantum computing, introducing the mathematical language of qubits, quantum gates, and unitary evolution that governs how information is processed in a quantum circuit. From there, we build toward Quantum Machine Learning, specifically Variational Quantum Circuits and data encoding strategies, which are the quantum components integrated into our hybrid models. Finally, we cover McNemar’s test, the statistical tool used throughout this thesis to determine whether any observed performance differences between models are genuinely meaningful rather than the result of chance.

2.1 Convolutional Neural Networks (CNNs)

Before deep learning became the dominant paradigm in computer vision, most pipelines relied on what are often called shallow learning systems. The underlying principle is as follows: the classifier was not expected to discover useful representations on its own. Instead, practitioners would hand-design feature extractors that pick up edges, corners, or texture patterns (for instance SIFT or HOG), and then pass these features to comparatively simple models such as Support Vector Machines (SVMs). This worked reasonably well in controlled settings. In practice, however, it proved insufficient when applied to images with greater visual complexity. Illumination changes, viewpoint shifts, deformations, and occlusions could easily break the feature design assumptions.

A major step away from this manual feature engineering paradigm came with the work of LeCun, Bottou, Bengio, and Haffner in their 1998 paper “Gradient-Based Learning Applied to Document Recognition” (LeCun et al., 1998). Their argument builds on an intuition borrowed from biology: visual processing is not done by looking at the entire scene at once, but by combining information from local receptive fields. CNNs operationalize this idea through convolution and weight sharing, which encourages locality and makes translation handling part of the architecture rather than something the model must learn from scratch via exhaustive data. In that paper, the LeNet-5 architecture became a clear example of how hierarchical features can be learned directly from pixels.

2.1.1 The Discrete Convolution

Let I denote the input image represented as a 2D grid of pixel values, and let K be a kernel (filter), i.e., a smaller matrix of learnable weights. Convolution’s objective is to produce an output feature map S by traversing the kernel to every position of the image and, at each position, it takes a weighted sum of that local image patch.

Table 2.1: Key Variables in the 2D Convolution Operation

Variable	Description
i, j	The spatial position (row, column) in the input I where the kernel is placed, and the corresponding output coordinate in S .
m, n	The kernel height (m) and width (n) of K .
u	Kernel row index in K , ranging from 0 to $m - 1$.
v	Kernel column index in K , ranging from 0 to $n - 1$.

For a 2D input I and a 2D kernel K of size $m \times n$, the output at spatial position (i, j) is:

$$S(i, j) = \sum_{u=0}^{m-1} \sum_{v=0}^{n-1} I(i+u, j+v) K(u, v)$$

The variables used in the convolution expression are summarized in Table 2.1.

Put differently, each output value $S(i, j)$ is obtained by multiplying each kernel weight $K(u, v)$ with the corresponding pixel $I(i+u, j+v)$ and then adding everything up. This local computation is also what people mean by *sparse connectivity* in CNNs: the value at a given output position depends only on a small neighborhood of the input (not the full image).

2.1.2 Structural Hyperparameters: Stride and Padding

We have two widely-known and practical hyperparameters which largely determine how convolution changes in the spatial setting: stride and padding.

- **Stride** (s): Stride controls how far the filter steps each time it moves across the input. With $s = 1$, the kernel shifts by one pixel and yields a relatively large output map. With $s = 2$, the kernel moves in larger jumps, so the output shrinks, or become smaller, accordingly (Karpathy, 2012).
- **Padding** (p): Adjacent to image boundaries, the kernel would otherwise extend beyond available pixels. Padding addresses this by adding extra pixels around the image border (typically zeros). This zero-padding is widely used to slow down spatial shrinkage in deeper CNNs and give the practitioner a better control for the output resolution.

Given an input size W_{in} , kernel size F , padding P , stride S , the output size W_{out} is:

$$W_{\text{out}} = \frac{W_{\text{in}} - F + 2P}{S} + 1 \quad (2.1)$$

2.1.3 The Importance of Activation Functions in Neural Networks

In the architecture of a Convolutional Neural Network, a convolution layer primarily performs linear operations (matrix multiplications and additions). If a network were composed exclusively of linear layers, the entire model would mathematically collapse into

a single linear transformation, regardless of its depth. Activation functions are therefore integrated after each linear layer to introduce non-linearity into the system. This allows the network to function as a universal approximator, capable of learning and representing the complex, non-linear morphological relationships inherent in histopathological images (Glorot et al., 2011; LeCun et al., 1998; Nair & Hinton, 2010).

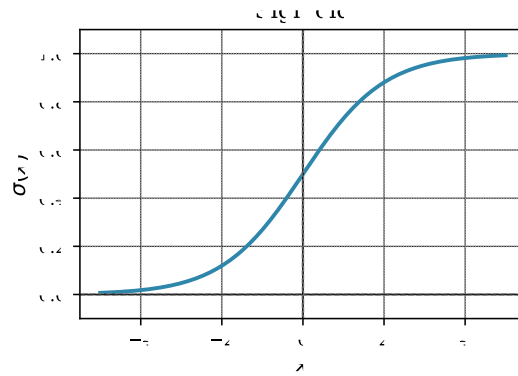
A summary of the activation functions evaluated in this study is provided in Table 2.2, followed by a visualization of their geometric properties in Figure 2.1.

As shown in Table 2.2, the four functions differ mainly in their output ranges. Sigmoid and Tanh both produce bounded outputs, between $(0, 1)$ and $(-1, 1)$ respectively, while ReLU and Leaky ReLU are unbounded on the positive side. The key difference between ReLU and Leaky ReLU is that Leaky ReLU allows a small negative output of $0.01x$ instead of zero, which helps prevent neurons from becoming permanently inactive during training.

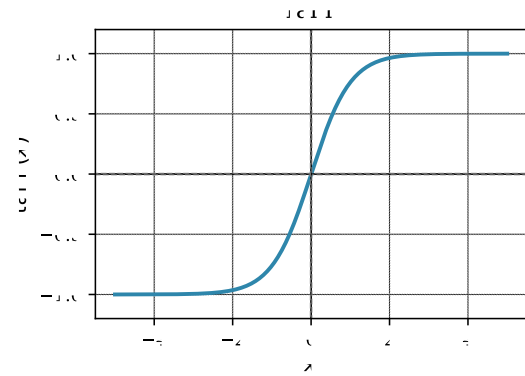
Table 2.2: Mathematical Definitions and Ranges of Common Activation Functions

Name	Formula	Range
Sigmoid	$\sigma(x) = \frac{1}{1+e^{-x}}$	$(0, 1)$
Tanh	$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$	$(-1, 1)$
ReLU	$f(x) = \max(0, x)$	$[0, \infty)$
Leaky ReLU	$f(x) = \max(0.01x, x)$	$(-\infty, \infty)$

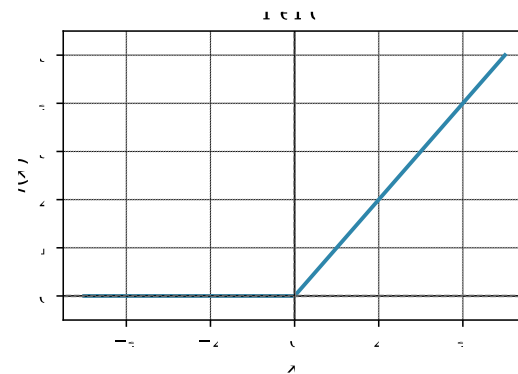
Figure 2.1 shows these differences visually. The Sigmoid and Tanh curves both flatten out at their extremes, which is where gradients become too small to drive learning effectively. ReLU avoids this by keeping a constant gradient for all positive inputs, though it outputs exactly zero for any negative value. Leaky ReLU looks nearly identical to ReLU but with a small visible slope on the left side of zero, ensuring that even neurons receiving negative inputs can still contribute to the learning process.



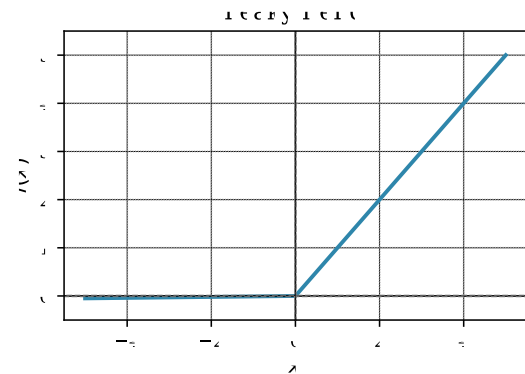
(a) Sigmoid



(b) Tanh



(c) ReLU



(d) Leaky ReLU

Figure 2.1: Visualization of common activation functions. (a) Sigmoid; (b) Tanh; (c) ReLU; (d) Leaky ReLU. These functions allow the model to capture non-linear features by thresholding or squashing the output of convolutional filters.

Sigmoid Activation Function. The Sigmoid function was historically one of the most widely used activation functions in early neural network development. It takes any real-valued input and "squashes" it into a range between 0 and 1, creating a smooth, S-shaped curve.

Sigmoid activation function's characteristics:

- **Probabilistic Interpretation:** Because the output is bounded between 0 and 1, it is particularly useful for models where the final output represents a probability, such as binary cancer classification.
- **Vanishing Gradient Problem:** A significant drawback is that for very high or very low input values, the function's gradient becomes nearly zero. This leads to the vanishing gradient problem, where weight updates during backpropagation become so small that the model stops learning.
- **Non-Zero Centered:** The output of the sigmoid is always positive, which can cause the gradients to zig-zag during optimization, potentially slowing down the convergence process.

Tanh (Hyperbolic Tangent) Function. The Tanh function is a mathematically shifted

version of the Sigmoid that maps input values to a range between -1 and 1 .

Tanh activation function's characteristics:

- **Zero-Centered Output:** Unlike the Sigmoid, Tanh is zero-centered. This generally makes the optimization process more stable and allows the network to converge faster during training.
- **Stronger Gradients:** The derivative of the Tanh function is steeper than that of the Sigmoid, which can lead to more significant weight updates in shallow networks.
- **Saturation Issues:** Similar to the Sigmoid, Tanh still suffers from the vanishing gradient problem in deep networks when inputs are far from zero, as the gradient effectively disappears at the function's tails.

ReLU (Rectified Linear Unit). ReLU has become the de facto standard for the hidden layers of modern CNN architectures. Its mathematical simplicity-outputting the input if it is positive and zero otherwise-is highly effective for deep learning tasks.

ReLU activation function's characteristics:

- **Computational Efficiency:** ReLU is exceptionally cheap to compute, requiring only a simple thresholding operation rather than expensive exponential calculations. This significantly reduces training time on large-scale datasets.
- **Sparsity:** By setting negative values to zero, ReLU encourages "sparse activation," where only a subset of neurons is active at any time. This mimicry of biological neurons helps the network learn more robust feature representations.
- **Linear Non-saturation:** Unlike squashing functions, ReLU does not saturate in the positive direction. This allows the gradient to flow freely through very deep networks, largely solving the vanishing gradient problem.
- **Dead ReLU Problem:** A major weakness is that if a neuron's weights are updated such that it always receives negative input, it will consistently output zero. This "dead" neuron stops contributing to the model and can no longer be updated via gradient descent.

Leaky ReLU. Leaky ReLU is a variant of the standard ReLU designed specifically to address the "dead neuron" issue. Instead of setting negative values to zero, it allows a small, non-zero gradient (typically $0.01x$) for negative inputs.

Leaky ReLU activation function's characteristics:

- **Gradient Continuity:** By providing a small slope for negative inputs, Leaky ReLU ensures that neurons continue to learn even when they are not highly activated.
- **Information Preservation:** It prevents the aggressive "dying" of neurons seen in standard ReLU, which can be beneficial when training on complex, high-resolution images where subtle negative signals might still contain useful diagnostic information.
- **Hyperparameter Dependency:** While effective, it introduces an additional hyper-

parameter (the slope of the leak) that may require manual tuning to achieve optimal performance in specific medical imaging contexts.

2.1.4 Pooling: Aggregation and Invariance

Following convolution and activation, many available CNNs include a pooling layer. Pooling reduces the spatial resolution of feature maps by combining activations over small neighborhoods, resulting in a more compact representation.

- **Max Pooling:** Max pooling keeps the largest activation within a local window (commonly 2×2). In effect, it preserves the strongest response in each region and is relatively insensitive to small shifts within the pooling window.
- **Average Pooling:** Average pooling replaces the window by its mean value. While it was used in earlier models such as LeNet-5, these days, max pooling has become increasingly prevalent in modern architectures because it tends to retain sharper, more discriminative signals if the image.

Pooling is typically used for two reasons:

- **Dimensionality Reduction:** It shrinks feature maps, which cuts down computation in later layers and can help limit overfitting.
- **Robustness:** If a feature shifts slightly within a pooling window, the pooled response is expected to still indicate the presence of that feature.

2.2 Fundamentals of Quantum Computing

To understand how quantum models work, it is first necessary to look at their basic building blocks. This section introduces the core concepts of quantum information, including how data is stored in qubits and how it is modified using quantum gates.

2.2.1 Qubits and Quantum States

The fundamental building block of quantum information processing is the qubit. While a classical bit is restricted to one of two discrete values - 0 or 1 - at any given moment, a qubit is a two-level quantum system that can exist in a more complex state space. Mathematically, the state of a qubit is represented as a unit vector within a two-dimensional complex Hilbert space, which allows for a much richer information-carrying capacity than its classical counterpart.

Dirac Notation and Basis States. In the field of quantum mechanics and quantum computing, it is standard practice to utilize **Dirac notation** (also known as bra-ket notation) to represent quantum states. The two primary states that correspond to the classical 0 and 1 are referred to as the computational basis states, denoted as $|0\rangle$ and $|1\rangle$.

The mathematical characteristics of these basis states include:

- **Vector Representation:** These states are defined as orthonormal column vectors in

the form:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

- **Orthonormality:** The basis is orthonormal, meaning the inner product of a state with itself is unity ($\langle 0|0\rangle = \langle 1|1\rangle = 1$), while the inner product between different states is zero ($\langle 0|1\rangle = 0$).
- **Standard Reference:** These states serve as the standard reference for all quantum operations and measurements performed within the circuits used in this study.

Superposition and the Born Rule. One of the most defining features of a qubit is its ability to exist in a **superposition**. This means that a pure single-qubit state $|\psi\rangle$ is not limited to being either $|0\rangle$ or $|1\rangle$, but can be expressed as a linear combination of both.

Superposition and measurement characteristics:

- **Linear Combination:** A general state is written as $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where α and β are complex numbers known as probability amplitudes.
- **Normalization Constraint:** To ensure the total probability of all possible outcomes remains equal to 1, these amplitudes must satisfy the normalization condition: $|\alpha|^2 + |\beta|^2 = 1$.
- **Born Rule and Measurement:** When a qubit in state $|\psi\rangle$ is measured in the computational basis, the superposition collapses. The outcome is 0 with a probability of $P(0) = |\alpha|^2$ and 1 with a probability of $P(1) = |\beta|^2$.
- **State Collapse:** Post-measurement, the qubit no longer sits in a superposition; it is forced into the observed basis state, effectively erasing the pre-measurement information contained in the amplitudes.

Geometric Representation: The Bloch Sphere. Because a pure single-qubit state is constrained by normalization and insensitive to an overall global phase, it can be visualized geometrically on the surface of a unit sphere known as the **Bloch Sphere**. This representation uses two real angles, θ and ϕ , to define the state:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

The Bloch Sphere characteristics:

- **Polar Angle (θ):** With a range of $0 \leq \theta \leq \pi$, this angle determines the "latitude" of the state. It dictates the relative weighting between $|0\rangle$ (North Pole, $\theta = 0$) and $|1\rangle$ (South Pole, $\theta = \pi$).
- **Azimuthal Angle (ϕ):** With a range of $0 \leq \phi < 2\pi$, this angle determines the "longitude" of the state. It represents the relative phase between the basis states, which is critical for interference effects in quantum algorithms.
- **The Z-Axis:** Any point on the Z-axis represents a state with no relative phase, while points on the equator of the sphere (where $\theta = \pi/2$) represent equal superpositions.

of $|0\rangle$ and $|1\rangle$.

- **Visualization Utility:** This geometric model is essential for understanding the action of quantum gates, such as R_y and R_x rotations, which physically move the state vector around the surface of the sphere.

2.2.2 Quantum Gates and Unitary Evolution

Classical logic gates (AND, OR, NOT) act on bits. Quantum circuits instead apply **quantum gates** to state vectors such as $|\psi\rangle$. Formally, a quantum gate corresponds to a unitary matrix. A matrix U is unitary when its conjugate transpose $U^\dagger$ is also its inverse:

$$U^\dagger U = U U^\dagger = I$$

This requirement ensures two things at once: the evolution is reversible, and the norm of the state vector is preserved (so total probability stays at 1) (Nielsen & Chuang, 2010). This matters in practice. If a gate were not unitary, the norm of the state vector would not be preserved, leading to inconsistent probability distributions.

The Pauli Group. Among the most common single-qubit operations are the Pauli matrices. They can be read as π -rotations about the principal axes of the Bloch sphere (MathWorks, 2025).

1. **Pauli-X (X):** Often described as the “quantum NOT” gate. It swaps the amplitudes of $|0\rangle$ and $|1\rangle$, corresponding to a rotation by π around the x -axis on the Bloch sphere.

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

2. **Pauli-Y (Y):** Effectively combines a bit flip with a phase flip.

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

3. **Pauli-Z (Z):** Commonly called the “phase-flip” gate. It keeps $|0\rangle$ as-is while multiplying $|1\rangle$ by -1 .

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

The Hadamard Gate. The **Hadamard** gate H is central because it turns a basis state into an equal-weight superposition:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H |0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle$$

$$H |1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$$

In many algorithms, H is the workhorse that sets up superposition so the circuit can act on many computational paths at once. In a QML setting, it is common to start with a layer of Hadamards to prepare an initial state that (in a loose sense) represents all inputs simultaneously (Nielsen & Chuang, 2010).

Parameterized Rotation Gates. Variational Quantum Circuits, which underpin many QML approaches, rely on gates that have trainable parameters. These parameters play a role similar to weights in a neural network and are adjusted by a classical optimizer (Tudisco et al., 2026). A simple choice is that we could use operations such as single-qubit rotations:

$$\begin{aligned} R_x(\theta) &= e^{-i\theta X/2} = \begin{bmatrix} \cos \frac{\theta}{2} & -i \sin \frac{\theta}{2} \\ -i \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{bmatrix} \\ R_y(\theta) &= e^{-i\theta Y/2} = \begin{bmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{bmatrix} \\ R_z(\theta) &= e^{-i\theta Z/2} = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix} \end{aligned}$$

By adjusting θ up or down during training processes, the optimizer steers the circuit through Hilbert space to reduce a pre-defined cost function. Analogously, in classical neural networks, parameters are updated iteratively to minimize a loss function.

2.3 Quantum Machine Learning (QML)

To bridge the gap between quantum theory and practical application, we must look at how Machine Learning works in a quantum environment. The following subsections detail the structure of variational circuits and the methods used to encode data for classification.

2.3.1 Variational Quantum Circuits (VQCs)

Variational Quantum Circuits, or we could use another terminology as Parameterised Quantum Circuits, are built upon a hybrid workflow: the quantum device will work with quantum states, while a classical optimizer will adjust a set of trainable parameters, according to a cost function. In practice, this setup ends up having a similar fashion to training a neural network, where parameters are updated iteratively to reduce a chosen loss function (Mitarai et al., 2018).

Anatomy of a VQC. Considering a standard VQC pipeline, typically, we have three stages, which are described as followed:

1. **Data Encoding:** A classical input x is mapped into a quantum state $|\psi_x\rangle$. This task is executed via an encoding unitary $U(x)$ operation. Details about encoding method will be discussed in Section 2.3.2.
2. **Parameterized Circuit (Ansatz):** A trainable unitary $U(\theta)$ is applied next. Here, θ denotes the set of parameters updated, or we could say them trainable weights, during the optimisation process.
3. **Measurement:** Finally, an observable M is measured on the resulting state to pro-

duce the model output $f(x, \theta)$, with which we can infer the results.

This computation is commonly written as:

$$\begin{aligned} |\phi(x, \theta)\rangle &= U(\theta)U(x) |0\rangle \\ f(x, \theta) &= \langle \phi(x, \theta) | M | \phi(x, \theta) \rangle \end{aligned}$$

In this notation, $f(x, \theta)$ represents the circuit's prediction for the classical input x given a set of trainable parameters θ . During the training process, this prediction is fed into a classical loss function, whose examples could be mean squared error or cross-entropy, and the optimizer uses that loss to decide how to update θ in the next iteration. The key point is that the quantum circuit support the forward pass (state preparation, transformation, measurement), where the pattern of the data, even with a complex pattern that cannot be easily seen if we use a simple Machine Learning approach, is expected to be explored by the quantum state. And in the meantime, the classical loop handles the parameter updates.

Overall, VQCs provide a practical way to combine quantum state manipulation with classical optimization, enabling hybrid learning algorithms on near-term quantum hardware (Mitarai et al., 2018).

2.3.2 Data Encoding

Data encoding, frequently referred to as quantum embedding, represents the critical interface between classical information and quantum computation. This stage is perhaps the most influential design choice in the development of a Quantum Machine Learning pipeline. It defines how a classical feature vector - such as the morphological characteristics extracted from a histopathological slide - is mapped into the high-dimensional Hilbert space of a quantum system. The choice of encoding is far from a mere formatting step; it fundamentally defines the expressivity of the quantum model and dictates which types of patterns the variational circuit can effectively perceive and process during the classification task.

Basis Encoding. Basis encoding is the most conceptually direct method of data loading, treating quantum states as a digital-to-quantum extension of classical bitstrings. In this scheme, a classical input is converted into a binary string of length n , which is then mapped to the corresponding computational basis state:

$$|x\rangle = |x_{n-1}x_{n-2} \dots x_0\rangle = |x_{n-1}\rangle \otimes |x_{n-2}\rangle \otimes \dots \otimes |x_0\rangle$$

Basis encoding method's characteristics:

- **Implementation Logic:** The state preparation relies on the application of Pauli- X gates to flip specific qubits into the $|1\rangle$ state, corresponding to the binary representation of the input vector.
- **Discrete Mapping:** It is highly effective for categorizing discrete data or integers, as it creates a rigid, one-to-one mapping between classical binary values and quantum basis vectors.

- **Superposition Deficit:** A significant limitation is the lack of inherent superposition during the loading phase; the resulting state is a single, deterministic basis vector rather than a complex probability distribution.
- **High Qubit Requirement:** For continuous-valued histopathology features, this method requires an extensive number of qubits to achieve sufficient numerical precision, which often exceeds the capacity of current quantum simulators.

Amplitude Encoding. Amplitude encoding is a mathematically sophisticated technique that stores classical data within the probability amplitudes of a quantum state. Given a normalized classical vector $x = [x_1, x_2, \dots, x_N]$, the quantum state is prepared as:

$$|\psi_x\rangle = \sum_{i=1}^N x_i |i\rangle$$

where $N = 2^n$ represents the total number of features and n is the number of qubits.

Amplitude encoding method's characteristics:

- **Exponential Density:** This scheme provides an exponential advantage in memory storage; a feature vector of size $N = 2^n$ can be encoded into only n qubits, making it the most compact representation for large-scale data.
- **Information Compression:** By utilizing the state amplitudes, the model can process massive datasets within a very small quantum register, which is theoretically ideal for high-resolution medical imaging.
- **Normalization Sensitivity:** The input data must be strictly normalized such that $\sum_{i=1}^N |x_i|^2 = 1$. While necessary for quantum consistency, this process can potentially erase the relative magnitude information between different tissue samples.
- **Circuit Depth Constraints:** Preparing an arbitrary amplitude-encoded state requires a complex sequence of controlled-rotation gates. On Noisy Intermediate-Scale Quantum (NISQ) devices, this often leads to high decoherence and gate errors before the actual processing begins.

Angle (Rotation) Encoding. Angle encoding, also known as qubit-rotation encoding, is the standard for modern hybrid classical-quantum architectures due to its high hardware efficiency. Each feature x_i is mapped directly to the rotation angle of a single-qubit gate, typically an R_y gate:

$$|x\rangle = \bigotimes_{i=1}^n R_y(x_i) |0\rangle$$

Angle encoding method's characteristics:

- **NISQ Robustness:** Because each feature is encoded into a single qubit using only one rotation gate, the state preparation is extremely shallow. This minimizes the time qubits spend in a fragile state, reducing the impact of quantum noise.
- **Trigonometric Non-linearity:** The encoding process inherently applies $\sin(x_i/2)$ and $\cos(x_i/2)$ functions to the input data. This creates a non-linear feature map

that allows the model to find separations in a higher-dimensional space that are not apparent in the raw pixel data.

- **Kernel-Like Representation:** This method effectively acts as a quantum feature map. Similar to the "kernel trick" in Support Vector Machines, it allows the hybrid model to identify complex correlations in tissue morphology.
- **Predictable Resource Scaling:** In this study, the number of qubits scales linearly with the number of features. This provides a stable and predictable resource requirement for the hybrid CNN-Quantum architecture, ensuring reliable benchmarking against classical baselines.

Chosen Embedding Method: Angle (Rotation) Encoding. For the purposes of this study, Angle Encoding was selected as the primary embedding strategy due to its optimal balance between maintaining the originality of the information from the CNN and computational efficiency. Basic encoding cannot be effectively utilized because the qubit cost associated with representing high-precision features from a CNN backbone is computationally prohibitive. While Amplitude Encoding could be employed and requires significantly fewer qubits, it represents the feature vector as a probability amplitude vector. This transformation represents the features in a way that differs fundamentally from what the feature vectors from the CNN backbone generally represent, potentially altering the underlying geometric relationships of the classical data. Thus, Angle Encoding was determined to be the most appropriate choice, as it maintains the integrity of the CNN-extracted information while remaining efficient enough to be executed on current quantum simulators.

2.3.3 Circuit Templates

To identify the optimal configuration for the variational quantum circuit (VQC), we evaluate three distinct circuit templates provided by the PennyLane framework. These templates differ in their entanglement strategies and parameter densities, allowing us to test how circuit expressibility impacts the classification of histopathological images.

Basic Entangler Layers

The *BasicEntanglerLayers* template is a computationally efficient architecture characterized by a simple entanglement scheme. Each layer consists of a series of single-qubit rotation gates (typically RX gates) followed by a cyclic sequence of CNOT gates that entangle adjacent qubits. This template is particularly useful for establishing a baseline performance, as it requires only one trainable parameter per qubit per layer, resulting in a lower computational overhead during the optimization process. As shown in Figure 2.2, the CNOT gates form a closed ring pattern, where each qubit is connected to its immediate neighbour, keeping the entanglement structure simple and the circuit shallow.

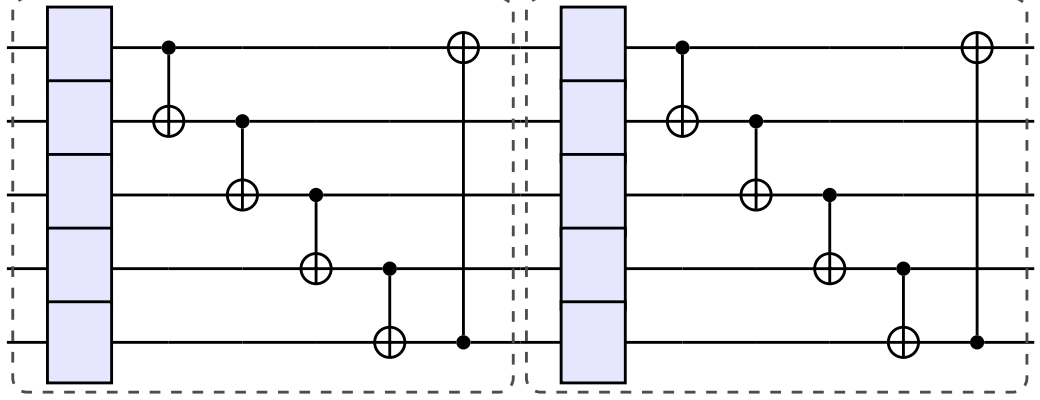


Figure 2.2: Visualization of the Basic Entangler Layers circuit template, consisting of single-qubit rotation gates on each qubit followed by a closed ring of CNOT gates. Adapted from (Xanadu AI, 2025a).

Strongly Entangling Layers

The *StronglyEntanglingLayers* template is designed to create high-degree correlations across the quantum register. Each layer utilizes the universal *Rot* gate, which consists of three trainable Euler angles (α, β, γ) per qubit. The entanglement is achieved through CNOT gates with a shifting "range" or jump between qubits in subsequent layers. This structure ensures that every qubit eventually interacts with every other qubit, providing the high expressibility needed to map complex features in high-dimensional Hilbert spaces. Figure 2.3 illustrates this structure, where each layer applies a full set of three-parameter rotation gates before the CNOT connections shift their range, gradually building long-range correlations across the entire qubit register.

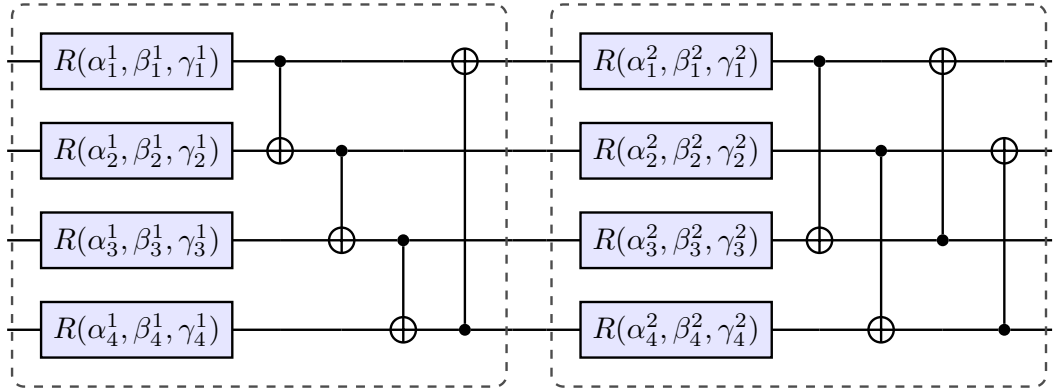


Figure 2.3: Visualization of the Strongly Entangling Layers circuit template, consisting of universal *Rot* gates on each qubit followed by CNOT gates with a shifting range between qubits across layers. Adapted from (Xanadu AI, 2025c).

Simplified Two-Design

The *SimplifiedTwoDesign* template is inspired by the concept of 2-designs, which are used to distribute quantum states uniformly across the Bloch sphere. This architecture consists of an initial layer of rotation gates, followed by alternating layers of entangling

CNOT gates and controlled-rotation gates. It is designed to provide a balance between circuit depth and expressivity, aiming to achieve the performance of a full 2-design with a more hardware-efficient parameterization. This makes it an ideal candidate for testing whether complex state-space coverage leads to better classification accuracy. As seen in Figure 2.4, the circuit alternates between Pauli-Y rotation blocks and controlled-Z entangling layers, producing a regular and structured pattern that covers the Bloch sphere more uniformly than the Basic Entangler while remaining shallower than the Strongly Entangling design.

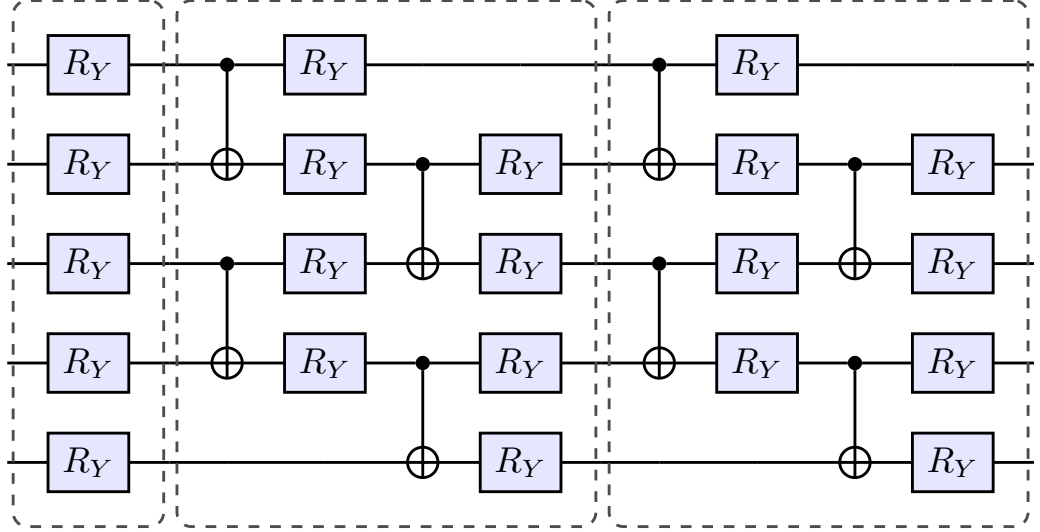


Figure 2.4: Visualization of the Simplified Two-Design circuit template, consisting of an initial layer of Pauli-Y rotation gates followed by alternating layers of controlled-Z entanglers and controlled-rotation gates. Adapted from (Xanadu AI, 2025b).

2.4 McNemar’s Test

McNemar’s test is a statistical test allowing us to two classifiers given the same set of test samples (Tudisco et al., 2026). Rather than just simply comparing the metric of accuracy, it asks a more precise question: when the two models disagree on a sample, is one model consistently right more often than the other?

2.4.1 Setup

Considering a test set and suppose we have two model A and B, we always have 4 values as follow: both models are correct, both are wrong, model A is correct and B is wrong, or model B is correct and A is wrong. These are recorded in a 2×2 contingency table (Table 2.3).

The cells a and d , where both models agree, are irrelevant to the comparison. The test focuses only on the discordant cells b and c , where the models give different answers.

Table 2.3: McNemar contingency table for comparing classifiers A and B.

	B Correct	B Incorrect
A Correct	a	b
A Incorrect	c	d

2.4.2 Test Statistic

The null hypothesis is that both models are wrong in equal proportion, i.e., $P(b) = P(c)$. The test statistic is:

$$\chi^2 = \frac{(b - c)^2}{b + c} \quad (2.2)$$

This follows a chi-squared distribution with the degree of freedom equal to one. If the p-value is calculated as below 0.05 ($\alpha = 0.05$), the null hypothesis is rejected and the difference between the two models is considered statistically significant.

It is worth noting that when the total number of discordant pairs ($b + c$) is small, a continuity-corrected version of the statistic is often recommended (Edwards, 1948):

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \quad (2.3)$$

However, since the test sets used in this thesis are sufficiently large, the standard form in Equation 2.2 is applied throughout.

2.4.3 Application in This Thesis

McNemar’s test is used in two places. In RQ3, it compares pairs of quantum configurations that differ in one hyperparameter at a time, to identify which design choices produce a statistically meaningful difference in classification behaviour. In RQ4, it is applied to each classical–quantum pair (same backbone, with and without the quantum layer) to confirm whether the observed accuracy and cancer recall improvements are statistically significant rather than due to chance.

3 Related Work

With the theoretical foundations in place, this chapter turns to the concrete building blocks and existing work that directly inform the design choices made in this thesis. We begin by surveying six well-known lightweight CNN families, covering their key architectural ideas and the specific variants selected for this study. We then review the most relevant hybrid quantum-classical approaches that have been applied to breast cancer detection, looking at how existing work combines classical feature extraction with quantum processing. The chapter closes by identifying the limitations that run through these studies, which is the gap this thesis sets out to address.

3.1 Current Well-Known Classical CNNs

Over the last decade, convolutional neural networks (CNNs) have undergone substantial advances in both accuracy and computational efficiency. In this section, we go over six widely used CNN families and the particular variants considered in this work:

- **MNASNet:** Tan et al. (2019) introduced a platform-aware Neural Architecture Search (NAS) method that explicitly takes device constraints into account. Rather than optimizing only for accuracy, MNASNet folds hardware latency into the reinforcement learning objective, which helps the final network line up with real-world mobile deployment requirements.
- **MobileNetV3:** Building on hardware-aware NAS, Howard et al. (2019) refined the MobileNet family to better fit practical resource limits. The variant having the name `mobilenetv3_small_100` used here is geared toward low-compute settings and relies on *h-swish* activations together with a $1.0\times$ width multiplier to keep a reasonable trade-off between speed and representation capacity.
- **EfficientNet:** Tan and Le (2019) proposed compound scaling, which jointly scales network depth, width, and input resolution in a balanced way. The “Lite” variants are even tailored for the use of mobile devices, mainly by removing operations that are less friendly to efficient deployment, such as the squeeze-and-excitation block.
- **RegNet:** Radosavovic et al. (2020) argued that it is often more useful to study a *design space* than to chase a single best model instance. By looking across populations of networks, they found that strong architectures tend to follow a simple “quantized-linear” rule. The `regnetx` and `regnety` variants mainly differ by whether squeeze-and-excitation modules are used.
- **GhostNet:** To cut down on computation spent on redundant feature maps, Han et al. (2020) proposed the GhostNet. The idea of this work is to generate additional “ghost” feature maps from a small set of intrinsic maps via inexpensive linear operations, reducing parameter count and computing resources needed while keeping accuracy mostly unchanged.
- **MobileViT:** Mehta and Rastegari (2022) moved beyond purely convolutional designs by introducing a lightweight hybrid that blends CNN components with Transformer-style global processing, the architecture that works remarkably in the field of Natural Language Processing. In effect, it treats Transformers as “convolutions” but in

a more compact form, aiming to capture global context while still benefiting from the spatial inductive bias and efficiency typically associated with CNNs.

Due to the constraints of the deployment and training resources considered in this thesis, one to two representative variants from each family are selected, with fewer than 5 million parameters. The chosen models, along with their parameter counts and single-item inference latency, are reported in Table 3.1.

Table 3.1: Parameter Count and Single-Item Inference Latency of Selected CNN Variants

Model Variant	Trainable Params (M)	Inference Time - Single (ms)
mobilenetv3_small_100	1.528	4.48
mobilevit_xs	1.937	7.29
regnetx_002	2.319	4.30
regnety_002	2.798	6.44
mnasnet_100	3.115	4.33
efficientnet_lite0	3.384	4.39
ghostnet_100	3.914	8.49

Note: The parameter counts and latency values were measured using the Pytorch framework within the timm library environment.

3.2 Hybrid Quantum-Classical Approaches on Breast Cancer

Recent work in quantum machine learning has pushed researchers to combine quantum circuits with classical deep learning, especially for serious medical imaging problems like breast cancer detection. Most of these setups follow a similar flow: a classical convolutional neural network (CNN) first extracts the image features, then a quantum model (often a quantum neural network: QNN, or a quantum convolutional neural network: QCNN) takes over to refine those features or carry out the final classification results. This hybrid split is practical because the CNN can handle high-dimensional image structure, while the quantum part is used, with the expectation of obtaining greater representational power with relatively few parameters.

Rifat et al. (2024) builds a hybrid pipeline that uses a modified ResNet50 backbone and an 8-qubit QNN for binary classification on the BreastMNIST dataset. Their key tweak is a lightweight convolutional layer inserted as a dimension-reduction step before the features are passed into the quantum module (which helps keep the quantum input manageable). With this design, they report an accuracy of 69%, which slightly improves on earlier hybrid baselines.

Matondo-Mvula and Elleithy (2024) take a more explicitly convolutional quantum route and propose a QCNN for breast cancer detection on the BreastMNIST ultrasound dataset. After encoding classical features into a quantum representation, their method applies 9-qubit entangled circuits that play the role of quantum convolutional layers. They achieve a validation accuracy of 87.17%, outperforming a classical CNN baseline and pointing to QCNN layers as a promising option when data are limited, which is common in medical imaging.

Tasnim et al. (2024) directly compare QCNN and CNN models using a subset of the

BreakHis histopathological image dataset. Their QCNN uses quantum convolution and pooling operations arranged over a 2x2 grid by using a 4-qubit kernel, with classical preprocessing handled upfront. The QCNN reaches 97% accuracy versus 82% for the classical CNN, and they also note faster convergence and a better F1 score, particularly when training with small batch sizes.

A comparative overview of the reviewed hybrid methodologies is provided in Table 3.2. As illustrated, there is a clear trend toward leveraging classical backbones like ResNet50 for initial feature extraction, while reserving quantum circuits - specifically Quantum Neural Networks (QNN) and Quantum Convolutional Neural Networks (QCNN) - for high-level feature refinement and classification. While the number of qubits remains limited, the significant performance gains reported suggest that quantum layers can enhance representational power and convergence rates.

Table 3.2: Detailed Comparison of Hybrid Quantum-Classical Approaches for Breast Cancer Detection

Paper	Dataset (Subsection)	Method Used	Qubits	Reported Results (Accuracy)	Key Feature / Innovation
Rifat et al. (2024)	BreastMNIST (780 full-image mammogram samples)	Modified ResNet50 + Conv layer (for dimension reduction) + QNN	8	69% (vs. 67% baseline)	Replaced fully connected layers with a convolutional layer for more efficient quantum input preparation.
Matondo-Mvula and Elleithy (2024)	BreastMNIST (Ultrasound dataset; 28x28 images)	QCNN (Quanvolutional Neural Network)	9	87.17% (vs. 83.33% classical CNN)	Uses angle embedding and 9-qubit entangled circuits with SU(4) gates to act as quantum filters.
Tasnim et al. (2024)	BreakHis (Histopathological images; binary subset)	QCNN (Quantum Convolution & Pooling)	4	97% (vs. 82% classical CNN)	Applied a 4-qubit kernel over a 2x2 grid; noted faster convergence and higher F1 scores in small-batch training.

3.3 Limitations of Existing Studies

The studies presented in Section 3.2 do show that hybrid quantum-classical models can work for breast cancer detection. However, several recurring limitations are evident across these studies.

A primary concern involves the nature of the data used for evaluation. Many approaches rely on simplified benchmarks, such as BreastMNIST, which lack the sample size and morphological variety inherent in real-world clinical imaging. Even when more complex datasets are utilized, evaluations often remain narrow; for instance, the BreakHis analysis in Tasnim et al. (2024) focuses only on a limited binary subset. Such restrictive data-processing practices fail to capture the full spectrum of clinical cases, making it difficult to determine if reported performance would remain stable across genuinely diverse images.

Furthermore, the architectural design of these models is frequently characterized by a lack of empirical justification. Most frameworks adopt a standard configuration — a classical CNN for feature extraction followed by a quantum classifier — yet the choice of the CNN backbone is often ad-hoc. Authors rarely provide a rationale for selecting a specific architecture or benchmark it against viable alternatives. For example, Rifat et al. (2024) utilizes ResNet50 without providing comparative experiments or selection criteria to support this choice. Consequently, it remains plausible that a different backbone could yield superior results within the same quantum configuration. A similar issue applies to the quantum component itself: key design parameters such as the number of qubits, circuit depth, and circuit template are typically fixed without any systematic search, leaving the question of whether the reported results reflect an optimal configuration largely unanswered.

Finally, among the three primary studies related to quantum-enhanced CNNs for medical imaging reviewed in this thesis (Matondo-Mvula & Elleithy, 2024; Rifat et al., 2024; Tasnim et al., 2024), none applies a formal statistical test to validate whether the quantum model is genuinely better than its classical counterpart. Reporting a higher accuracy number alone does not confirm that the difference is meaningful rather than due to chance, which limits the strength of the conclusions that can be drawn.

Table 3.3 summarises these limitations across the reviewed studies and positions the contributions of this thesis in relation to them. As shown in the comparison, existing literature focuses primarily on small-scale datasets and binary classification, often overlooking architectural optimization and statistical significance—gaps that this thesis aims to bridge.

Table 3.3: Comparison of existing hybrid quantum-classical studies and this thesis across key methodological dimensions.

Paper	Dataset Used	Class. Type	CNN Search	Quantum Search	Statistical Test
Rifat et al. (2024)	BreastMNIST (780 mammogram samples)	Binary	×	×	×
Matondo-Mvula and Elleithy (2024)	BreastMNIST (780 ultrasound images; 28×28)	Binary	×	×	×
Tasnim et al. (2024)	BreakHis (binary subset only)	Binary	×	×	×
This Thesis	Full BreakHis (7,909 images)	Multi-class (8 classes)	✓	✓	✓

To address these limitations, our work utilizes the more challenging full BreakHis dataset and conducts a comprehensive experimental analysis across multiple dimensions. Specifically, we perform systematic comparisons involving:

- various CNN architectures to identify the most effective feature extractor;
- diverse quantum configuration settings to optimize the quantum classifier; and
- paired evaluations of quantum and classical approaches sharing identical CNN backbones, validated using McNemar’s test.

This rigorous methodology provides the empirical evidence necessary to inform the optimal selection of a CNN backbone for our hybrid model, ultimately enhancing both the robustness and the generalizability of our research outcomes.

4 Study Design

This chapter describes how the research is actually carried out. We begin by stating the overall goal and the four research questions that break it down into testable steps. From there, we introduce the BreakHis dataset and walk through the evaluation settings, covering data preprocessing, augmentation choices, and the hardware and framework setup used throughout all experiments. The chapter then explains the specific approach taken for each research question, from selecting the best classical backbone, to tuning the quantum hyperparameters, to determining which design choices genuinely matter, and finally to comparing classical and hybrid models head to head.

4.1 Experimental Workflow

This thesis investigates whether the integration of a quantum component to a breast cancer histopathology image classifier produces measurable improvements in classification performance. Prior work on hybrid quantum-classical approaches for breast cancer classification (Matondo-Mvula & Elleithy, 2024; Rifat et al., 2024; Tasnim et al., 2024) has typically followed a straightforward pattern: propose an architecture that combines a CNN with a quantum layer, then compare it against its purely classical counterpart. While this establishes a proof of concept, it leaves several practical questions unanswered. Which CNN backbone is best suited as the classical component? Which quantum design choices actually drive performance gains? Are the observed improvements statistically meaningful?

To address these gaps, this thesis proposes a more comprehensive empirical study structured as a sequential workflow. Rather than evaluating a single fixed architecture, we first identify the most suitable CNN backbone across several candidates, then systematically explore the space of quantum design choices to find the best configuration, and finally perform a rigorous head-to-head comparison between the classical and quantum-enhanced versions of each backbone. Each stage directly informs the next, ensuring that design decisions are grounded in evidence rather than assumption.

According to the four research questions defined in Chapter 1, we conduct four corresponding experimental steps:

- **Classical Backbone Selection:** In this step, we conduct a systematic evaluation of various CNN architectures. This allows us to identify the most effective feature extractor for the hybrid experiments by comparing models based on their accuracy and the time required for training.
- **Quantum Hyperparameter Optimization:** Once the classical backbone is fixed, we integrate a Variational Quantum Circuit (VQC) and perform an exhaustive grid search. We iterate through combinations of qubits, circuit depths, and circuit templates to find the configuration that maximizes classification performance.
- **Influence of Quantum Hyperparameters:** We analyze the variance across our grid search results to determine the sensitivity of the hybrid model. This step isolates which specific quantum design parameters have the most significant statistical impact on the final accuracy.

- **Classical vs. Quantum Performance:** In the final stage, we perform a head-to-head comparison between the best classical model and the optimized hybrid model. We apply the McNemar test to the results to ensure that any observed performance differences are statistically significant rather than the result of chance.

In response to these questions, all models are compared under the same benchmarking setup, ensuring fair results.

4.2 Dataset Description and Replication Package

This section describes the dataset used throughout all experiments in this thesis, followed by a description of the replication package that allows the full study to be reproduced independently.

4.2.1 Dataset Description

We work with the Breast Cancer Histopathological Image Classification (BreakHis) dataset (Spanhol et al., 2015). It contains 7,909 microscopy images of breast tumor tissue from 82 patients, captured at four magnification factors ($40\times$, $100\times$, $200\times$, and $400\times$). The images are RGB with a resolution of 700×460 pixels, 8-bit depth per channel, and stored in PNG format. In detail, the dataset includes 2,480 benign and 5,429 malignant samples. The data was put together in collaboration with the P&D Laboratory – Pathological Anatomy and Cytopathology, Paraná, Brazil¹.

Regarding the labels: each tumor type is split into four histological sub-classes:

- **Benign:** Adenosis (A), Fibroadenoma (F), Phyllodes Tumor (PT), Tubular Adenoma (TA)
- **Malignant:** Ductal Carcinoma (DC), Lobular Carcinoma (LC), Mucinous Carcinoma (MC), Papillary Carcinoma (PC)

4.2.2 Replication Package

To support transparency and reproducibility, all source code, training scripts, and evaluation pipelines used in this thesis are publicly available at a GitHub repository (Vo, 2026).

The repository is implemented in Python and includes the following components:

- `src/` — core source code covering dataset preparation, model training for both classical and hybrid quantum-classical architectures, and all evaluation routines;
- `tests/` — scripts for running hypothesis testing and model benchmarking, corresponding directly to the statistical analyses reported in RQ3 and RQ4;
- `requirements.txt` — a full list of dependencies with pinned versions to ensure a consistent software environment;

¹<http://www.prevencaoediagnose.com.br>

- `Makefile` — a set of commands to reproduce each stage of the study, from dataset download and model training to backbone timing benchmarks, model comparison, and McNemar statistical testing.

The repository is structured so that each research question in this thesis maps directly to a corresponding evaluation command. For instance, backbone timing results (RQ1) can be reproduced via `make evaluate-backbone-timing`, quantum hyperparameter results (RQ2 and RQ3) via `make paired-hypothesis-test`, and the classical versus quantum comparison (RQ4) via `make mcnemar-test`. Full instructions for setting up the environment and running all experiments are provided in the repository’s `README.md` file.

4.3 Evaluation Settings

To ensure the experiments are accurate and consistent, we must define the specific settings used for testing. This section explains how the data was prepared and describes the hardware and software used to train the models.

4.3.1 Data Preprocessing

The dataset is split into training, validation, and test subsets using a 50%-25%-25% split. Since the classes are imbalanced, data augmentation was applied exclusively to the training set to increase the effective size of minority-class samples and improve model generalisation.

Augmentation Techniques Applied:

- **Horizontal and Vertical Flips:** Unlike everyday photographs, histopathology slides do not have a fixed “correct” orientation. A tissue sample looks the same and carries the same diagnostic meaning whether it is flipped horizontally or vertically. This makes flipping a safe and effective way to generate additional training examples without changing their labels (Tellez et al., 2019).
- **Rotations (up to 90°):** For the same reason, rotating tissue images does not alter the underlying biology. Structures such as cell nuclei and glandular patterns look diagnostically identical at different angles, so rotations naturally expand the training data while keeping it realistic and meaningful (Komura & Ishikawa, 2018).

Why Some Techniques Were Left Out:

- **Colour Jittering:** In histopathology, colour is not just visual, it carries clinical meaning. The purple and pink tones produced by Haematoxylin and Eosin (H&E) staining help both pathologists and models identify different tissue types. Randomly changing these colours during training could confuse the model by making meaningful colour cues appear unreliable (Tellez et al., 2019).
- **Elastic Deformations:** This technique stretches and warps the image in ways that do not occur naturally in tissue samples. While it is sometimes used in other medical imaging tasks, applying it here risks creating unrealistic images that do not

reflect real histopathology slides, which could ultimately hurt model performance.

All images are subsequently resized to 224×224 pixels to conform to standard CNN input requirements.

The resulting class distribution for training (before and after augmentation), validation, and test is listed in Table 4.1.

Table 4.1: Number of sample images in each class: original training set (before augmentation) and after 50/25/25 split with balanced augmentation.

Class	Train (Original)	Train (Augmented)	Validation	Test
Adenosis	222	1725	111	111
Fibroadenoma	507	1725	253	254
Phyllodes Tumor	226	1725	113	113
Tubular Adenoma	285	1725	143	142
Ductal Carcinoma	1725	1725	863	863
Lobular Carcinoma	313	1725	156	157
Mucinous Carcinoma	396	1725	198	198
Papillary Carcinoma	280	1725	140	140
Total	3954	13800	1977	1978

4.3.2 Framework and Hardware Configuration

We built the hybrid CNN-quantum pipeline using PennyLane ² together with PyTorch ³. All quantum circuits were executed using PennyLane’s state-vector simulator, meaning no real quantum hardware was used at any point in this study. This choice is standard practice in hybrid quantum-classical machine learning research, as current quantum hardware suffers from noise and limited qubit counts that would introduce confounding factors beyond the scope of this thesis. All experiments are run on an NVIDIA V100 GPU with 32GB virtual RAM, which is sufficient to accommodate all the chosen backbone architectures and keeps training time manageable.

For reproducibility, a fixed random seed of 42 was used across all experiments, covering model weight initialization, dataset splitting, and data augmentation. Each experiment was run once under the same controlled conditions, as the deterministic setup ensures that results are fully reproducible without requiring multiple runs.

Regarding training hyperparameters, all models were trained for 20 epochs, which was found to be sufficient for convergence across all tested backbones without signs of overfitting on the validation set. A batch size of 32 was selected after empirical testing, as it produced the most stable training process while remaining within the memory constraints of the GPU for all model configurations, including the more memory-intensive hybrid quantum models. The Adam optimizer was used with a learning rate of 0.01. All backbones are initialized with pretrained ImageNet weights through transfer learning, meaning the feature extraction layers are already well-trained and only the classification head needs to adapt to the BreakHis dataset. In this case, a higher learning rate is beneficial because it allows the model to take larger update steps and reach a good solution faster, which is

²<https://pennylane.ai/>

³<https://pytorch.org>

particularly important given the tight budget of only 20 training epochs. Using a lower rate such as 0.001 was also tested but resulted in slower convergence, with the model not fully adapting within the same number of epochs and showing no improvement in final accuracy.

4.4 Approaches for Each Research Question

To keep the comparisons across research questions fair, we follow one shared experimental pipeline for training and evaluation (see Section 4.3). This entails applying identical dataset preprocessing, including augmentation steps and train/validation/test split strategy, and also the same implementation environment for every experiment.

Figure 4.1 sums up the overall study design and shows how each research question fits into the full workflow. The dataset is first split into training (50%), validation (25%), and test (25%) sets, with the training set then balanced. The process branches into two parallel tracks: a Training Time Benchmark to identify the fastest CNN backbone using dummy tensors, and RQ1 to evaluate baseline accuracy across all backbones. The fastest backbone is carried forward into RQ2 to optimize quantum configurations. RQ3 then identifies which quantum design choices significantly affect accuracy through McNemar tests, and RQ4 compares classical and quantum performance across all backbones. The arrows in the figure show the dependencies between each step, ending with the final results and conclusions.

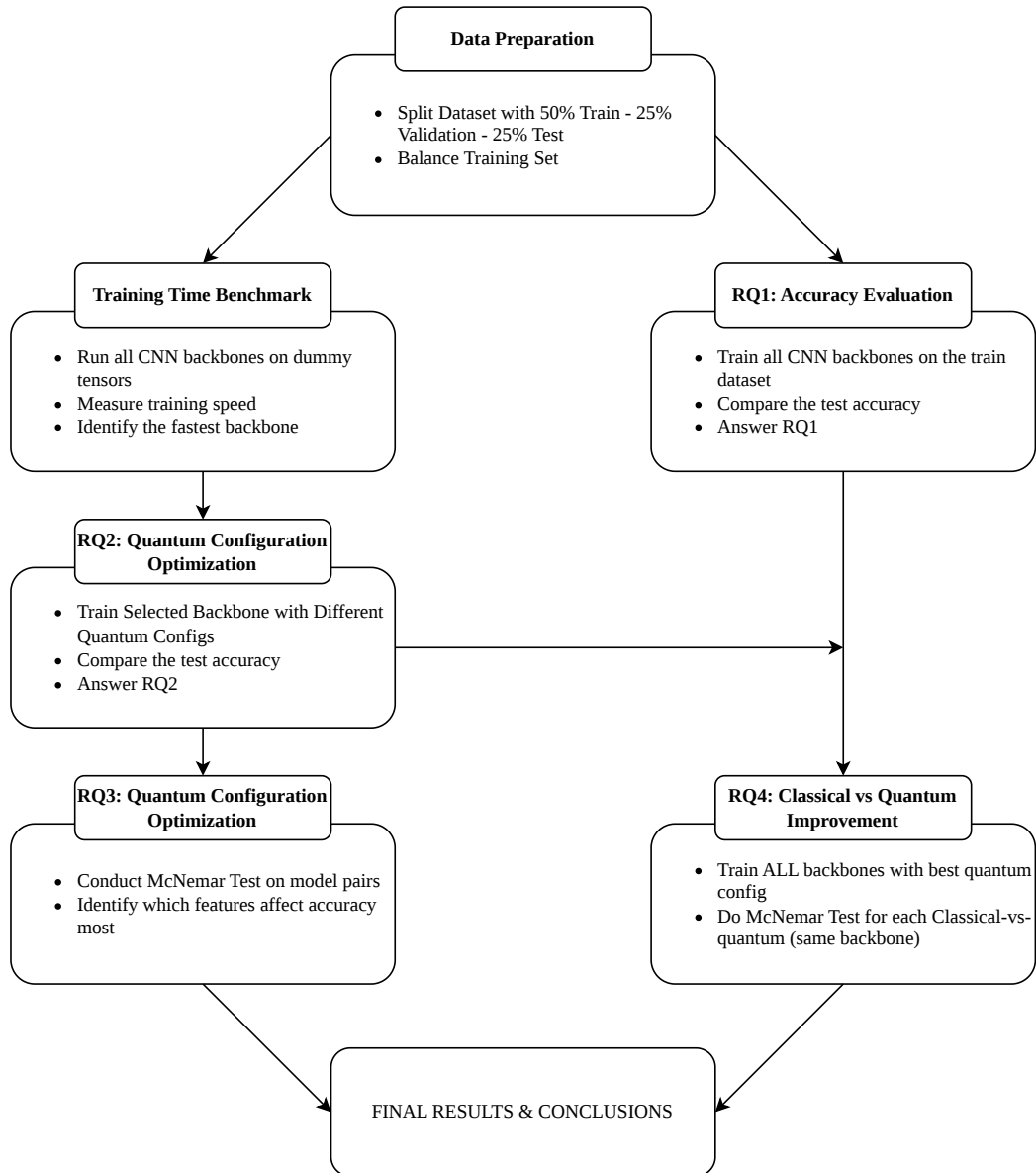


Figure 4.1: Overview of the four-stage study design workflow.

Figure 4.2 shows the full pipeline from the classical CNN backbone through the angle encoding layer, the variational quantum circuit with its configurable hyperparameters, and the final classification head. A classical CNN backbone extracts image features, which are then angle-encoded into a qubit register of 8 or 16 qubits. A Variational Quantum Circuit (VQC) processes the quantum state using one of three circuit templates (BasicEntanglerLayers, StronglyEntanglingLayers, SimplifiedTwoDesign) at a chosen depth (3 or 5 layers), yielding 12 configurations in total. Pauli-Z measurements on the first 8 qubits produce an 8-dimensional output vector, which is passed to a dense classification head for 8-class breast cancer subtype prediction.

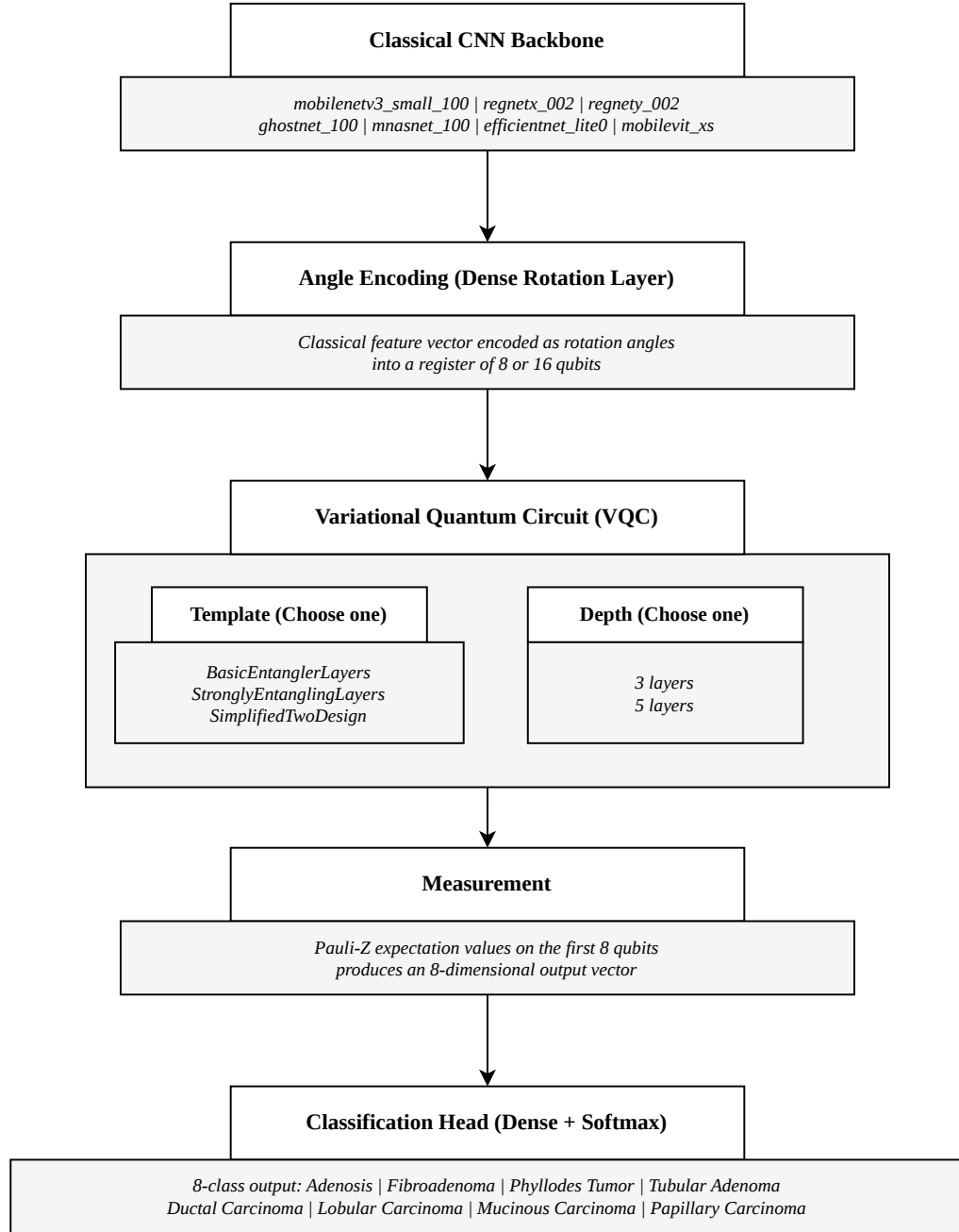


Figure 4.2: Overview of the hybrid quantum-classical architecture.

For RQ1, we start by benchmarking a set of state-of-the-art CNN backbones, including the variants listed in Table 3.1. The objective of this stage is to establish a rigorous classical baseline for each model before we bring quantum layers into these model. Each candidate's backbone is trained on the balanced training split and evaluated on the 25% test set. For this stage, test accuracy and training time are the main metrics, since they let us line up the backbones and see which one performs best for breast cancer image classification.

From there, RQ2 shifts the focus from selecting a backbone to examining the quantum

part of the hybrid model. Quantum circuit simulation is computationally intensive and can incur substantially longer training times than purely classical pipelines, so before running a full evaluation, we first carry out a Training Time Benchmark using dummy tensors. This step identifies the most computationally efficient backbone from the earlier pool, which we then carry forward into the quantum experiments.

Regarding the setup in RQ2, the selected backbone is used for a grid search over these following quantum configurations:

- **Number of Qubits:** We test configurations with 6 and 8 qubits.
- **Circuit Templates:** We try three variational circuit families:
 - `BasicEntanglerLayers`: One-parameter single-qubit rotations on each qubit, followed by a closed chain (ring) of CNOT gates.
 - `StronglyEntanglingLayers`: Single-qubit rotations and entanglers, following the circuit-centric classifier approach, whose details are described by Schuld et al. (2020).
 - `SimplifiedTwoDesign`: A simplified 2-design made up of Pauli-Y rotations and controlled-Z entanglers, as proposed by Cerezo, Sone, et al. (2021).
- **Layer Depth:** Each circuit template is evaluated at depths of 3 and 5 layers.

We then take the configuration with the highest test accuracy and treat it as the "best quantum config" for the later research questions.

Building on that, RQ3 looks more closely at which quantum design choices actually matter. Instead of only reporting accuracy differences, we run McNemar tests between model pairs to check whether changes in specific quantum parameters lead to statistically meaningful shifts in classification performance. This helps narrow down which components in the hybrid architecture are genuinely driving improvements.

Finally, RQ4 compares each classical backbone to its quantum-enhanced version. All backbones are retrained using the single best quantum configuration found in RQ2, ensuring a controlled and internally consistent comparison. For each backbone, we apply the McNemar test to the classical-vs-quantum pair to determine whether the quantum augmentation produces a statistically significant improvement in classification accuracy.

5 Results

This chapter presents the results of the four research questions, following the experimental setup described in the previous chapter. We begin with RQ1, which identifies the best classical backbone in terms of both accuracy and training efficiency. We then move to RQ2, where we search for the optimal quantum circuit configuration on top of the selected backbone. RQ3 follows with a closer look at which quantum hyperparameters actually drive meaningful differences in classification performance, using McNemar’s test to separate real effects from noise. The chapter closes with RQ4, which puts the best classical and hybrid models head to head across all backbones to determine whether the quantum layer delivers a consistent and statistically significant improvement.

Throughout this chapter, the two primary evaluation metrics are overall accuracy and malignant cancer recall. This choice is deliberate and clinically motivated. In a breast cancer diagnosis setting, the cost of a false negative - missing a malignant case - is far greater than the cost of a false positive. Recall on the malignant class therefore captures the most safety-critical aspect of model performance: how well the model detects cancer when it is actually present. Overall accuracy complements this by reflecting the model’s general ability to correctly classify all eight tumor subtypes. While additional metrics such as precision, F1-score, and confusion matrices would provide a more complete picture, they are reserved for future work where a more detailed per-class breakdown is the primary focus. For the purposes of this study, which is concerned with whether the quantum layer produces a consistent and statistically significant improvement over its classical counterpart, accuracy and recall together are sufficient to answer that question in a clear and clinically meaningful way.

5.1 RQ1: Classical Backbone Selection

The classification performance of all CNN backbone architectures, which has been evaluated on the BreakHis dataset, is presented in Table 5.1. Every model ran for 20 epochs on the augmented training set (see Section 4.3) before being evaluated on the separated test split.

Among all candidates, the variant `regnetx_002` achieved the highest test accuracy with a test accuracy of 90.80%, which is followed closely by the two backbones `mnasnet_100` and `efficientnet_lite0`, both reaching 90.34%. `ghostnet_100` placed in the middle tier with a test accuracy of 89.74%, while the variant `regnety_002` achieved 87.16%. `mobilenetv3_small_100` reached 86.30%, and `mobilevit_xs` had a noticeable lower accuracy of 75.38%.

Among the stronger models, training accuracy ranged from 91.67% to 97.88%, meaning most of them had no real trouble fitting the training data. The bigger question, then, is generalization, and that is where the gaps between models actually show up. Notably, the high training accuracies - several models exceeding 97% - compared to their lower test accuracies suggest a degree of overfitting, even with data augmentation applied. For instance, `regnetx_002` reached 97.88% training accuracy but only 90.80% on the test set, a gap of over 7 percentage points. This pattern appears across most backbones and indicates that while the models fit the training data well, generalizing to unseen samples remains a challenge.

Table 5.1: Classification accuracy (%) and loss across training, validation, and test sets for each CNN backbone.

Model	Train Acc	Train Loss	Val Acc	Val Loss	Test Acc	Test Loss
mnasnet_100	97.70	1.9217	90.34	1.3703	90.34	1.3717
efficientnet_lite0	97.72	1.9218	89.63	1.3776	90.34	1.3695
regnetx_002	97.88	1.9208	89.43	1.3814	90.80	1.3656
ghostnet_100	97.28	1.9231	88.11	1.3919	89.74	1.3734
regnety_002	96.83	1.9270	87.20	1.4037	87.16	1.4043
mobilenetv3_small_100	97.32	0.4842	86.24	0.6630	86.30	0.6592
mobilevit_xs	91.67	1.9534	74.46	1.5270	75.38	1.5238

Regarding the training time benchmark, we have Table 5.2 that reports these figures, where we employed dummy tensors. `mobilenetv3_small_100` recorded the fastest training epoch time at 25.11 seconds (0.42 min) with the lowest memory footprint of 27.94 MB. `regnetx_002` was close behind at 25.81 seconds per epoch. On the other end, `mobilevit_xs` struggled here: it was the slowest at 55.98 seconds per epoch and logged the highest batch inference time at 146.74 ms.

While `regnetx_002` led on accuracy, `mobilenetv3_small_100` makes a stronger case when efficiency matters. Its 86.30% test accuracy (Table 5.1) is competitive, and its training speed and memory footprint are superior to all other candidates. Consequently, it was selected as the backbone for the quantum hyperparameter search in RQ2.

Table 5.2: Training time benchmark results for each CNN backbone measured using dummy tensors.

Model	Params (M)	Inf. Single (ms)	Inf. Batch (ms)	Inf./Sample (ms)	Epoch (sec)	Epoch (min)	Mem. (MB)
mobilenetv3_small_100	1.528	4.48	18.61	0.073	25.11	0.42	27.94
regnetx_002	2.319	4.30	32.89	0.128	25.81	0.43	32.10
regnety_002	2.798	6.44	35.52	0.139	26.51	0.44	33.94
ghostnet_100	3.914	8.49	57.10	0.223	30.59	0.51	40.56
mnasnet_100	3.115	4.33	60.20	0.235	30.61	0.51	28.69
efficientnet_lite0	3.384	4.39	72.29	0.282	37.08	0.62	43.22
mobilevit_xs	1.937	7.29	146.74	0.573	55.98	0.93	40.63

Summary for RQ₁

`regnetx_002` achieved the highest test accuracy at 90.80% among all classical backbones, but it was not the final choice for quantum experiments. `mobilenetv3_small_100`, otherwise, was selected as the backbone for quantum experiments due to its superior training efficiency (25.11 sec/epoch, 27.94 MB memory) combined with a sufficiently competitive test accuracy of 86.30%.

5.2 RQ2: Quantum Hyperparameter Optimization

Table 5.3 summarises how all twelve quantum circuit configurations performed on top of the selected backbone `mobilenetv3_small_100`. These configurations were tested by combining the three hyperparameters: circuit template (`two_design`, `strong`, `basic`), number of qubits (8 and 16), and circuit depth (3 and 5 layers).

The `two_design` template seemed to consistently outperform the other two templates across all qubit and depth combinations, with the top-performing configuration was `Two Design` with depth 3 and 8 qubits, achieving a test accuracy of 92.72% and a validation

accuracy of 92.11%, which also represents an improvement of 6.42 percentage points over the classical-only baseline of 86.30%. This was followed by the combination of `Two Design` with depth 5 and 8 qubits, at 92.57%. Interestingly, an increase in the number of qubits from 8 to 16 within the `two_design` template led to a slight accuracy drop, with test scores of 91.96% (depth 5) and 90.65% (depth 3), suggesting that additional qubits do not provide a noticeable improvement for this task.

Regarding the template `strong`, it produced intermediate results, with test accuracies between 69.21% and 86.30%.

About the template `basic`, it exhibited markedly weaker performance, with test accuracies ranging from 52.12% to 80.64%, and several of its configurations could not even match the classical baseline. The average test accuracy by template was 91.97% for `two_design`, 81.53% for `strong`, and 67.88% for `basic`.

Table 5.3: Test and validation accuracy (%) for all quantum circuit configurations evaluated on `mobilenetv3_small_100`.

Configuration	Train Acc (%)	Val Acc (%)	Test Acc (%)	Train Loss	Test Loss
Two Design (Depth 3, 8 Qubits)	99.33	92.11	92.72	0.0114	0.2564
Two Design (Depth 5, 8 Qubits)	99.37	91.60	92.57	0.0116	0.2772
Two Design (Depth 5, 16 Qubits)	98.84	91.50	91.96	0.0179	0.2659
Two Design (Depth 3, 16 Qubits)	98.88	89.18	90.65	0.0198	0.3051
Strong (Depth 5, 16 Qubits)	96.94	86.44	85.54	0.0883	0.5177
Strong (Depth 3, 8 Qubits)	97.22	85.84	86.30	0.1077	0.5451
Strong (Depth 5, 8 Qubits)	97.11	84.42	85.09	0.1128	0.5882
Basic (Depth 5, 8 Qubits)	96.38	80.32	80.64	0.0932	0.7894
Basic (Depth 5, 16 Qubits)	92.71	78.15	79.27	0.1911	0.7567
Strong (Depth 3, 16 Qubits)	87.96	66.21	69.21	0.3162	1.0433
Basic (Depth 3, 16 Qubits)	82.11	58.93	59.50	0.5944	1.4298
Basic (Depth 3, 8 Qubits)	87.22	51.75	52.12	0.1209	0.9204

Summary for RQ₂

The best quantum configuration is the `Two Design` circuit template with depth 3 and 8 qubits, achieving a test accuracy of 92.72% on top of `mobilenetv3_small_100`. The circuit template appeared to be the dominant factor in performance, with `two_design` outperforming `strong` and `basic` across all settings.

5.3 RQ3: Influence of Quantum Hyperparameters

In order to assess which quantum hyperparameters significantly affect classification accuracy on the test set, McNemar’s test was used to make pairwise comparisons across all 12 quantum configurations, which were trained on top of `mobilenetv3_small_100` in Section 5.2. Since the test evaluates whether two classifiers make statistically different errors on the same set of samples, it is highly suitable for comparing model variants that differ in a single design choice. For each hyperparameter, only pairs of configurations that differ in that specific parameter while holding all other parameters constant are selected to be compared, ensuring that observed effects can be attributed to that parameter only.

For the **template** hyperparameter, there are 3 possible pairwise combinations (`basic` vs. `strong`, `basic` vs. `two_design`, `strong` vs. `two_design`), each of which is

evaluated across 4 conditions (2 qubit settings $\times$ 2 depth settings), eventually yielding 12 total comparisons. For **depth**, each template is evaluated at both depth levels for every qubit setting, resulting in a total of 3 templates $\times$ 2 qubit settings = 6 comparisons. Similarly, for **qubits**, the two settings of 8 and 16 qubits are evaluated with each template at each depth level, yielding 3 templates $\times$ 2 depth settings = 6 comparisons. In total, we have 24 pairwise McNemar tests that were conducted.

Referring to the Table 5.4, the number of statistically significant comparisons ($p < 0.05$) for each hyperparameter is summarised. Out of the 24 total tests, 11 showed statistical significance. The template turned out to be the hyperparameter that had the clearest impact, having 7 out of 12 comparisons (58.3%) reaching significance. This was followed by the parameter Depth, which showed moderate influence - 3 out of 6 comparisons (50.0%) being significant. The hyperparameter qubits, in the meantime, demonstrated the weakest effect, with only 1 out of 6 comparisons (16.7%) reaching significance.

Table 5.4: Number of statistically significant McNemar test results ($p < 0.05$) per hyperparameter across all pairwise comparisons.

Hyperparameter	Significant Tests	Total Tests
Template	7 (58.3%)	12
Depth	3 (50.0%)	6
Qubits	1 (16.7%)	6
Total	11	24

Table 5.5 presents the significant pairwise comparisons for the hyperparameter template. The `two_design` template appeared to be the strongest circuit design in general, winning in 5 out of 7 significant comparisons. The `strong` template consistently underperformed, and notably at depth 5 with 16 qubits, the comparisons involving `strong` reached extreme significance ($p < 0.0000005$ against `basic`, and $p < 0.0000002$ against `two_design`).

Interestingly, in two comparisons under 16-qubit settings, `basic` outperformed the two other templates, which indicated that the template ranking is not fixed and can shift depending on the qubit count.

Table 5.5: Statistically significant pairwise McNemar test results for the **Template** hyperparameter ($p < 0.05$). Non-significant comparisons are omitted.

Comparison	Qubits	Depth	p-value	Winner
basic vs. strong	8	3	0.0146	basic
basic vs. two_design	8	5	0.0040	two_design
strong vs. two_design	8	5	0.00002	two_design
basic vs. two_design	16	3	0.0192	basic
strong vs. two_design	16	3	0.0401	two_design
basic vs. strong	16	5	0.0000005	basic
strong vs. two_design	16	5	0.0000002	two_design

Table 5.6 covers the significant comparisons for the depth hyperparameter, and the depth's effects seem to be strongly template-dependent.

For the `two_design` template, depth 5 outperformed depth 3 at both qubit settings ($p = 0.0014$ at 8 qubits, $p = 0.0031$ at 16 qubits), indicating that deeper circuits are beneficial

for this design. For the `strong` template, the opposite was true, since at 16 qubits, depth 3 significantly outperformed depth 5 ($p = 0.00006$), which suggests that increasing depth can be detrimental in certain configurations. No significant depth effect was observed for the `basic` template.

Table 5.6: Statistically significant pairwise McNemar test results for the **Depth** hyperparameter ($p < 0.05$). Non-significant comparisons are omitted.

Template	Qubits	p-value	Winner
two_design	8	0.0014	depth 5
two_design	16	0.0031	depth 5
strong	16	0.00006	depth 3

For qubits, only one comparison reached statistical significance: the `strong` template at depth 3, where 16 qubits outperformed 8 qubits ($p = 0.0095$). Other templates and depths did not witness any significant difference between 8 and 16 qubits, from which we could infer that 8 qubits are sufficient in most configurations and that scaling to 16 qubits provides no general benefit.

Summary for RQ₃

The circuit template is the most impactful quantum hyperparameter, demonstrating statistically significant effects in seven of the twelve comparisons (58.3%). The `two_design` template consistently surpasses `strong` and `basic`. Depth has a moderate effect, with three out of six comparisons (50.0%) and interacts with the template; specifically, deeper circuits are advantageous for `two_design` but detrimental for `strong`. The number of qubits has the least influence, only one out of six comparisons (16.7%), and in the majority of configurations, the performance of 8 qubits was comparable to that of 16.

5.4 RQ4: Classical vs. Quantum Performance

To evaluate whether adding a quantum layer produces a statistically meaningful improvement, all seven CNN backbones were retrained with the best quantum configuration identified in RQ2 (`two_design`, depth 3, 8 qubits). Using the same test set of 1,978 samples, McNemar’s test was then applied to each classical–quantum pair. This test set included 1,358 malignant samples, which were used to evaluate cancer recall.

Table 5.7 reports accuracy delta, cancer recall delta, and McNemar p-values for both metrics across all backbones. All seven backbones demonstrated improved accuracy following the addition of the quantum layer, and every improvement was statistically significant ($p < 0.001$) for overall accuracy.

The largest accuracy gain was observed for `mobilevit_xs`, which improved by 18.35 percentage points (from 75.38% to 93.73%), with a McNemar p-value below 1×10^{-6} . `regnety_002` followed with a gain of 7.13 per cent (from 87.16% to 94.29%), and the backbone `mobilenetv3_small_100` improved by 6.42 per cent (from 86.30% to 92.72%). The smallest improvement was recorded for `mnasnet_100` at 2.83 per cent, which was still statistically significant ($p = 0.000020$).

For cancer recall specifically, Table 5.8 reports the McNemar contingency values for each

Table 5.7: Classical vs. quantum performance comparison for all backbone models. Accuracy delta and cancer recall delta are reported in percentage points (pp). McNemar’s test is applied separately for overall accuracy and malignant cancer recall on the same 1,978 test samples (1,358 malignant).

Backbone	Acc Delta (pp)	McNemar p (Acc)	Recall Delta (pp)	McNemar p (Recall)
mobilevit_xs	+18.35	< 0.000001	+2.72	< 0.000001
regnety_002	+7.13	< 0.000001	+1.69	0.000078
mobilenetv3_small_100	+6.42	< 0.000001	+0.88	0.059230
ghostnet_100	+5.06	< 0.000001	+0.66	0.066457
efficientnet_lite0	+3.64	< 0.000001	+0.81	0.029096
regnetx_002	+3.19	< 0.000001	+0.44	0.238593
mnasnet_100	+2.83	0.000020	+0.66	0.052345

backbone, showing how many samples changed classification outcome between the classical and quantum models. Three out of seven backbones showed statistically significant improvements in cancer recall ($p < 0.05$): mobilevit_xs, regnety_002, and efficientnet_lite0. The remaining four backbones showed positive recall deltas that did not reach statistical significance. For mobilevit_xs, the quantum model correctly reclassified 44 malignant samples that the classical model had missed, while only 7 were flipped in the opposite direction. A similar pattern, though smaller in magnitude, was observed for regnety_002 (27 gained vs. 4 lost) and mobilenetv3_small_100 (23 gained vs. 11 lost).

Table 5.8: McNemar contingency table values for cancer recall across all backbone models.

“Gained” refers to malignant samples correctly classified by the quantum model but not by the classical model; “Lost” refers to the reverse.

Backbone	Both Correct	Both Wrong	Gained	Lost	p-value (Recall)
mobilevit_xs	1300	7	44	7	< 0.000001
regnety_002	1320	7	27	4	0.000078
mobilenetv3_small_100	1316	8	23	11	0.059230
ghostnet_100	1336	3	14	5	0.066457
efficientnet_lite0	1327	10	16	5	0.029096
regnetx_002	1336	4	12	6	0.238593
mnasnet_100	1336	5	13	4	0.052345

Summary for RQ₄

The incorporation of the quantum layer yielded statistically significant accuracy enhancements across all seven backbones, with p-values less than 0.001 for overall accuracy. Cancer recall exhibited statistically significant improvement for three of the seven backbones (mobilevit_xs, regnety_002, and efficientnet_lite0), while the remaining four showed positive but non-significant recall deltas. As we could expect, the largest gains were observed for weaker classical baselines: mobilevit_xs improved by 18.35 per cent in accuracy and 2.72 per cent in cancer recall. Even the strongest classical backbone, regnetx_002, benefited from a statistically significant improvement of 3.19 per cent in accuracy, indicating that the quantum enhancement is both consistent and not confined to less capable models.

6 Discussions

With the results established, this chapter turns to what they actually mean. We begin by interpreting the findings across all four research questions, situating them within the broader literature and drawing out the key patterns that emerge. From there, we consider the practical implications for those working on clinical or applied medical imaging systems, followed by a discussion of what the findings contribute to the research community. The chapter closes by identifying the most promising directions for future work that follow naturally from this study's findings and limitations.

6.1 Interpretation of Results

The findings across all four research questions provide consistent evidence that: layering a quantum component on top of a classical CNN backbone actually improves breast cancer histopathology classification in ways that are measurable and statistically meaningful, and this is seen even when using realistic and difficult datasets. What follows is an interpretation of these results, both for the specific questions asked and for how they fit into the broader literature.

For RQ1, the narrow accuracy range across most backbones (86–91%) is itself a meaningful finding. When models start with ImageNet pretrained weights, the feature extraction layers are already well-tuned before training even begins. This reduces architectural differences and shifts the competition mainly to the classification head. The clear outlier, `mobilevit_xs` at 75.38%, is worth examining more carefully. Despite being a more modern architecture, it underperformed every pure CNN backbone. This underperformance can likely be attributed to a mismatch in inductive bias. CNNs are built around locality and translation invariance, which naturally match how diagnostic patterns are structured in tissue images. Attention mechanisms, by contrast, need substantially more data per class to learn meaningful relationships across the image, and the class imbalance in BreakHis likely put `mobilevit_xs` at a disadvantage that augmentation alone could not fix.

For RQ2, the `two_design` circuit template won out across all qubit and depth settings. This circuit uses Pauli-Y rotations and controlled-Z gates, and it beat the more complex `StronglyEntanglingLayers` template in every configuration. This initially appears counterintuitive, since a more complex circuit should be able to learn more. This could be attributed to "barren plateaus", a known problem in quantum circuit training where gradients become too small to be useful as circuits get deeper or wider (Cerezo, Sone, et al., 2021). The simpler `two_design` circuit avoids this by keeping the training problem more manageable. This is not a paradox but a known tradeoff: a circuit that is theoretically more expressive can be harder to train in practice than a shallower, more structured one.

For RQ3, the most important finding is that circuit template choice drove meaningful differences far more often than qubit count. This challenges a common assumption in the QML literature, where adding more qubits is often treated as the main way to improve performance. The data here suggests the opposite priority: getting the circuit structure right matters more than scaling qubit resources. This finding is consistent with the known barren plateau problem. More qubits increase the circuit's theoretical capacity, but they

also make optimization harder, which may explain why 8 and 16 qubit configurations performed similarly in most cases. The interaction between depth and template type is also informative. Deeper circuits helped `two_design` but hurt `strong`. This means depth is not an independent setting that can be increased freely. Its effect depends entirely on how the circuit handles entanglement. For `two_design`, extra layers expand state coverage in a controlled way. For `strong`, they appear to push the circuit into a regime where gradients become unreliable.

For RQ4, the key result is the quantum layer significantly improved accuracy across all seven tested backbones. Notably, this improvement was not confined to architecturally weaker baselines, even strong baselines such as `regnetx_002` (+3.19 per cent) and `efficientnet_lite0` (+3.64 per cent) even improved, alongside more substantial gains in architecturally weaker models such as `mobilevit_xs` (+18.35 per cent). This may indicate that the quantum layer provides a complementary computational contribution that classical layers alone cannot replicate. The cancer recall results corroborate this interpretation: three out of seven backbones improved significantly in detecting malignant samples, which is the most clinically important metric. However, these results were obtained using classical quantum simulation via PennyLane rather than real quantum hardware. Since simulators do not reproduce the noise, gate errors, and decoherence of physical devices, the gains reported here reflect the potential of the architecture as a mathematical model rather than its current performance on real processors (Preskill, 2018). Validation on actual quantum hardware remains a necessary next step.

Taken together, these results constitute a positive proof-of-concept for quantum-enhanced CNN classification on a realistic histopathological dataset, directly addressing the gap in the literature identified in Section 3.

6.2 Implications for Practitioners

The findings carry a few practical implications worth considering for anyone working on clinical or applied medical imaging systems.

The most immediate practical implication is that quantum layers, when set up correctly, can meaningfully improve both overall accuracy and cancer recall without requiring a complete rebuild of an existing pipeline. In the hybrid architecture evaluated here, the quantum component sits between the classical feature extractor and the final classification head. This means practitioners can keep a well-validated CNN backbone and simply add a quantum layer on top, rather than replacing the whole model. Given that regulatory and validation workflows in medical AI tend to be expensive and slow, an additive integration strategy of this kind reduces the clinical validation burden substantially.

Having said that, the results also show that quantum configuration choices are not straightforward. The performance gap between the best (`two_design`, 92.72%) and worst (`basic`, 52.12%) configurations is substantial, or in fact, even bigger than the gap observed between classical backbone architectures. This means that deploying a hybrid model without a careful systematic hyperparameter selection could produce a system that actually performs even worse than the classical baseline. Quantum hyperparameter optimisation should therefore be treated as a mandatory step, rather than an auxiliary consideration. In particular, circuit template selection appears to matter more than simply scaling up to more qubits, since the latter showed the weakest influence on accuracy in

this study.

On the hardware side, all experiments were run using quantum circuit simulation on classical hardware (an NVIDIA V100 GPU). Within the scope of this study, simulation on a V100 GPU was computationally feasible, but the inference times for quantum-augmented models are considerably higher than for purely classical ones. Anyone considering deployment in latency-sensitive settings - such as real-time intraoperative pathology - should benchmark end-to-end inference times carefully before adopting this architecture for time-sensitive applications. Access to real quantum hardware may eventually reduce simulation overhead, though near-term devices remain noisy and error-prone, which introduces additional engineering challenges that fall outside the scope of this thesis.

6.3 Implications for Researchers

For the research community, this study contributes several findings that extend beyond a direct performance comparison.

First, it shows that a fair and rigorous comparison between hybrid quantum-classical models and classical baselines requires more than just reporting accuracy numbers. Using McNemar’s test on the same held-out samples makes it possible to tell apart genuine improvements from noise - and the results show that raw accuracy differences can be misleading. A 2.83 per cent improvement (`mnasnet_100`) can be statistically significant for overall accuracy but not for the clinically relevant recall metric. Researchers reporting hybrid model results on medical imaging tasks should routinely include paired statistical tests alongside point estimates, not treat them as optional.

Second, the finding that circuit template choice matters more than qubit count pushes back on a common assumption in the QML literature, where scaling up qubit resources is often presented as the main way to improve performance. This study suggests that architectural decisions within the quantum component deserve just as much attention as qubit resource scaling.

Finally, this study used the full BreakHis dataset with proper class balancing techniques and implemented an 8-class classification setting. Several prior studies worked with binary classification on smaller, more simplified subsets (Matondo-Mvula & Elleithy, 2024; Rifat et al., 2024; Tasnim et al., 2024), which makes results harder to generalize. The findings here suggest that quantum-enhanced models can perform well in a more realistic and demanding setting. Future work should follow the same standard and avoid simplifying evaluation conditions in ways that may inflate reported performance metrics.

6.4 Future Work

The study of this thesis opens up several directions for future work, each following naturally from its findings and limitations.

- **Backbone-specific quantum tuning.** This study applied the same quantum configuration across all backbones, but that would not necessarily be the best approach. Since RQ3 showed clear interactions between circuit template and other hyperparameters, each backbone might perform better with its own tailored quantum

setup. A joint search over both backbone and quantum hyperparameters, using something like Bayesian optimisation to keep the search manageable, would give a much clearer picture of the hybrid design space.

- **More circuit templates and encoding strategies.** Only three ansatz families were tested here (`BasicEntanglerLayers`, `StronglyEntanglingLayers`, and `SimplifiedTwoDesign`), all using angle encoding. Other encoding methods, such as amplitude encoding, might interact differently with the classical feature vectors and could either improve results or reveal new weaknesses. A systematic comparison of encoding strategies under the same evaluation setup would be a very useful follow-up.
- **Testing on real quantum hardware.** Simulation is a reasonable starting point, but things could behave differently in real quantum devices. Running the hybrid pipeline on physical quantum hardware is a necessary step to understand whether these results hold in practice. Noise-aware training, error mitigation, and hardware-efficient circuit designs all deserve attention here. The `two_design` template's shallower depth might make it more resilient to hardware noise.
- **Broader clinical evaluation.** Future work should report per-subtype results across all eight BreakHis tumor classes, include precision and F1 alongside recall, and ideally test on data from different institutions or imaging platforms to check how well the model generalizes. Adding explainability methods to understand what drives the model's predictions would also help build clinical trust.
- **Reducing computational cost.** The quantum layer adds meaningful overhead to both training and inference. Exploring lighter quantum circuit designs that preserve accuracy gains without the simulation cost, and eventually leveraging real quantum hardware as it matures, are both worthwhile long-term goals.

7 Threats to Validity

Like any empirical study, this work is subject to a set of limitations that warrant explicit acknowledgment. This chapter outlines the key boundaries across four dimensions of threats to validity as defined by Wohlin et al. (2012): conclusion, internal, construct, and external validity. Keeping these in mind helps ensure the findings are interpreted in the right context and with an appropriate level of confidence.

Conclusion validity. This category concerns the reliability of the statistical relationship between the treatment and the outcome. In this study, although a fixed random seed of 42 was used to make results deterministic, each experiment was only run once due to the high computational cost of simulation. Ideally, using different random seeds and averaging the results would provide a more robust statistical picture and ensure that the observed quantum benefits are not due to an exceptionally favorable initialization.

Internal validity. All quantum experiments during the hyperparameter search phase (RQ2 and RQ3) were run using a single classical backbone (`mobilenetv3_small_100`), chosen for its training speed. This means the optimal quantum setup found may not generalize perfectly to every other backbone. Additionally, all models were trained with a fixed learning rate (0.01) and optimiser (Adam) without per-model tuning, which may put specific architectures at a disadvantage. Finally, the quantum search only covered a small grid of qubit counts and circuit templates to keep training time manageable, potentially missing more effective configurations.

Construct validity. This refers to the extent to which the experiment actually measures the concepts it claims to. While overall accuracy and malignant recall were used, they do not capture the full clinical picture (e.g., precision and F1 score). Furthermore, a significant threat to construct validity is the use of classical quantum simulation via PennyLane rather than real quantum hardware. Simulators do not reproduce the environmental noise, gate errors, and decoherence found in real devices. Current quantum processors operate in the Noisy Intermediate-Scale Quantum (NISQ) era, where gate errors and short coherence times fundamentally limit circuit depth (Preskill, 2018). In practice, this means that a variational circuit of depth 3 or 5 executed on real hardware could produce substantially degraded outputs compared to its simulated counterpart, potentially reversing or diminishing the accuracy gains reported in this thesis. Therefore, the "quantum advantage" observed here measures the potential of the mathematical model rather than its current performance on physical quantum processors, and results should be interpreted accordingly until validation on real hardware becomes feasible.

External validity. The primary threat to generalization is the use of the BreakHis dataset, which originates from a single hospital and covers only 82 patients. While a strong benchmark, it may not reflect the full variation in staining methods, scanner types, and patient demographics seen globally. Consequently, these results serve as a proof-of-concept for hybrid quantum-classical architectures rather than a tool ready for immediate clinical deployment.

8 Conclusion

In summary, this thesis explored whether adding a quantum layer to a classical CNN pipeline could produce measurable improvements in breast cancer histopathology classification, using the BreakHis dataset as a realistic and challenging benchmark. The study was implemented around four research questions, each aiming at a different aspect of the hybrid quantum-classical design.

For RQ1, seven lightweight CNN backbones were benchmarked. Most performed in a similar range, with test accuracies between 86% and 91%. `regnetx_002` achieved the highest test accuracy with a test accuracy of 90.80%, while `mobilenetv3_small_100` was chosen for the quantum experiments because of its faster training speed and competitive accuracy.

For RQ2, twelve quantum circuit configuration combinations were tested by varying the circuit template, qubit count, and circuit depth. The `SimplifiedTwoDesign` template with 8 qubits and depth 3 outperformed all others, reaching a test accuracy of 92.72%, which is a 6.42 percentage point gain over the classical-only baseline.

For RQ3, McNemar’s test was applied across 24 pairwise comparisons to identify which quantum hyperparameters actually are the factors driving statistically significant differences. Circuit template was the most impactful factor, with 58.3% of its comparisons reaching significance, followed by depth at 50.0% and finally qubit count at just 16.7%. This suggests that choosing the right ansatz matters more than simply adding more qubits.

For RQ4, each classical backbone was compared against its quantum-enhanced version under the same evaluation setup. The quantum layer produced statistically significant accuracy improvements across all seven backbones, and meaningfully improved cancer recall for three out of seven. These gains were consistent regardless of how strong the baseline was. Even the best classical model (`regnetx_002`) achieved a statistically significant accuracy improvement of 3.19 percentage points in accuracy and 0.44 percentage points in malignant recall.

Taken together, these results offer a solid proof-of-concept for quantum-enhanced breast cancer classification on a realistic dataset, addressing a clear gap in prior work that largely relied on simplified benchmarks. The findings also give practical guidance for future hybrid model design: prioritise circuit template selection over qubit scaling, and always pair accuracy comparisons with statistical tests to determine whether observed differences are statistically meaningful.

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